A FRAMWORK OF INTEGRATED SD-WAN WITH HONEYPOT AND IOT MONITRING

Publications

1. SAFIE-ELDIN N. MOHAMED, RAGHDA R. G. SHAFIQ, A. ESSAM, A. YASER, ESRAA A. ALGOHARY, E. OSMAN, M. AZAB, M. MOSBEH, S. HALAWA, WAFAA F. MOHAMMED,"**SD-WAN and Cowrie Honeypot Synergy in Modern Networks - From Traffic Optimization to Threat Trapping**," submitted to journal of network and Management system, (IF= 4.1).
2. SAFIE-ELDIN N. MOHAMED, RAGHDA R. G. SHAFIQ, A. ESSAM, A. YASER, ESRAA A. ALGOHARY, E. OSMAN, M. AZAB, M. MOSBEH, S. HALAWA, WAFAA F. MOHAMMED, " **Integrated SD-WAN Framework with Cowrie Honeypot and Zabbix for Network Orchestration, Threat Detection, and IoT Monitoring**," submitted to Scientific Reports, (IF= 3.8).

Introduction



Let’s start with EVE-NG

What: Industry-standard tool for virtual lab environments.

Why: Emulate complex networks without hardware.

**Key Use: Test SD-WAN, security,
and multi-vendor setups risk-free.**



SD-WAN

What: Software-defined WAN for centralized, application-aware networking.

Why: Replace rigid MPLS with agile, cost-effective hybrid links (broadband, LTE).

Key Use: Optimize performance, automate policies, and embed security.

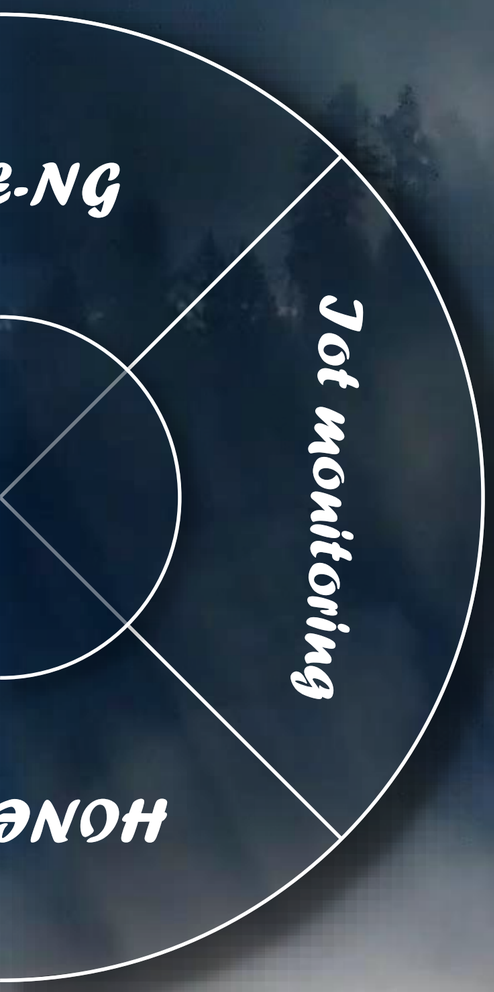


Cowrie Honeypot

What: Open-source SSH/Telnet honeypot based on linux.

Why: Lure attackers, log tactics, and gather threat intelligence.

Key Use: Detect brute-force attack & expose lateral movement attempts.



Iot monitoring

What: Tracking connected devices (sensors, cameras, smart systems) to ensure they work reliably and securely.

Why:
Prevent Failures , Save Costs,
Boost Security , Optimize Data.

Key Use: Real-Time LAN Health Monitoring,
Cost-Efficient Scalability,

Proposed model 1

SD-WAN

Agility & Scalability

-Spin up new sites in minutes, not months. Adapt to cloud/IoT growth

-Impact: Deploy 100 sites as easily as 1

objective

Agility & Scalability

-Spin up new sites in minutes, not months. Adapt to cloud/IoT growth

-Impact: Deploy 100 sites as easily as 1

Reduce Human Error

-Automate configurations with zero-touch provisioning

-impact: Cut misconfigurations by 62%

Cost Efficiency

-Replace expensive MPLS with affordable broadband/5G links

-impact: Cut WAN costs by 75–86%

Centralized Security

-Integrate firewalls, encryption, and threat intelligence at the edge

-Impact: Block attacks before they enter the network

Performance Optimization

-Dynamically route traffic based on app priority, latency, packet loss

-Impact: Ensure VoIP/video never drops

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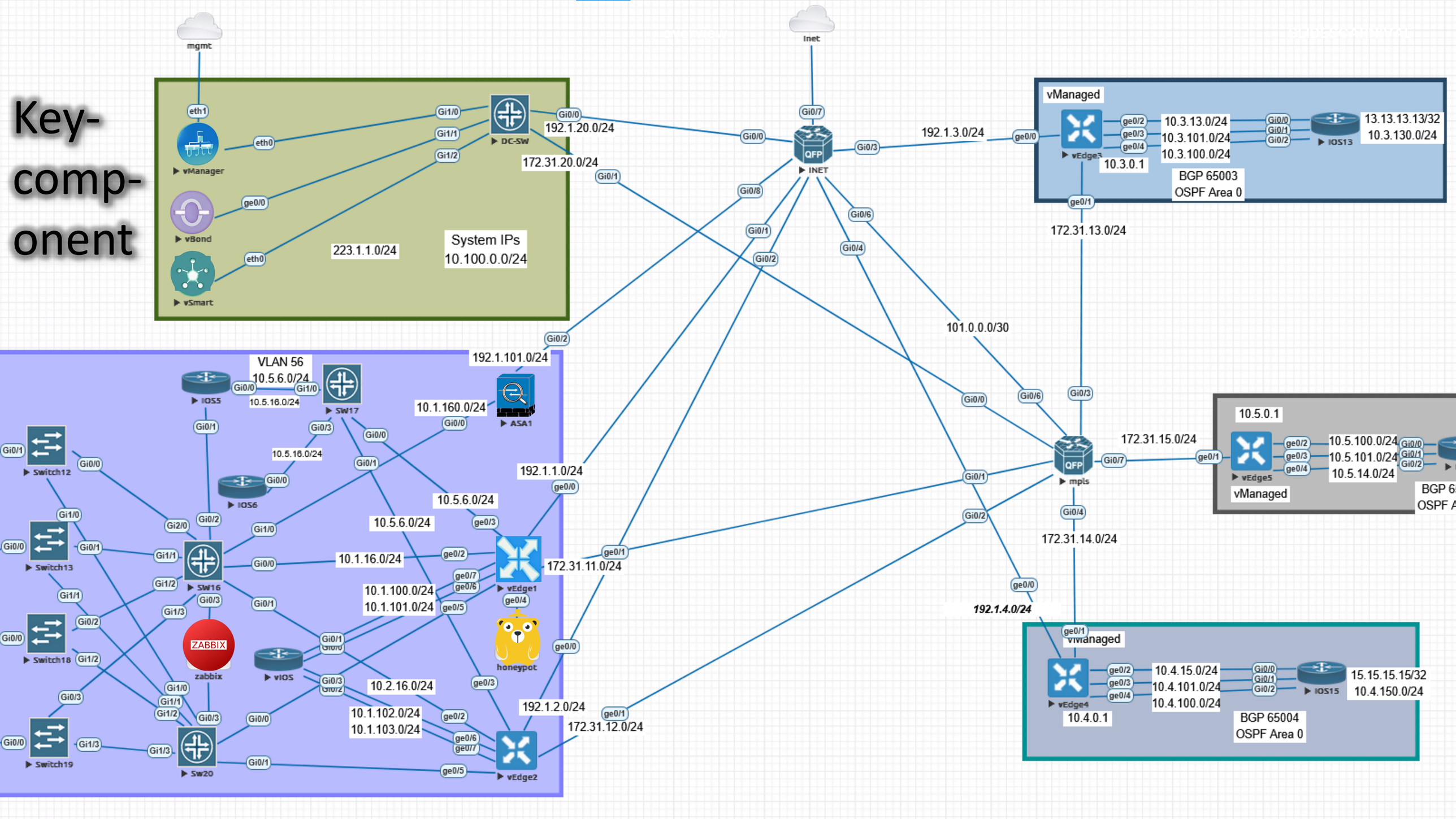
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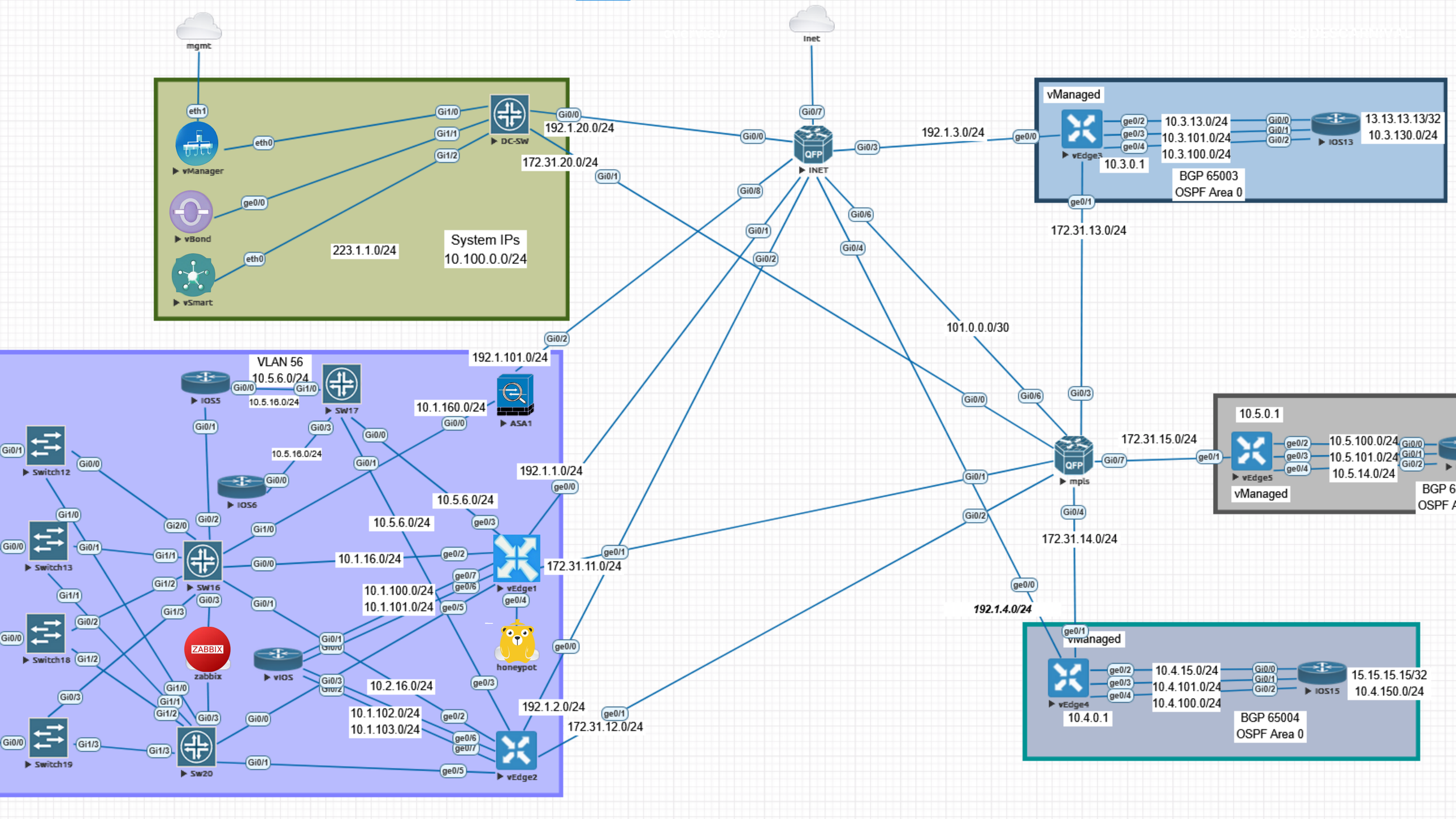
Key-comp-onent





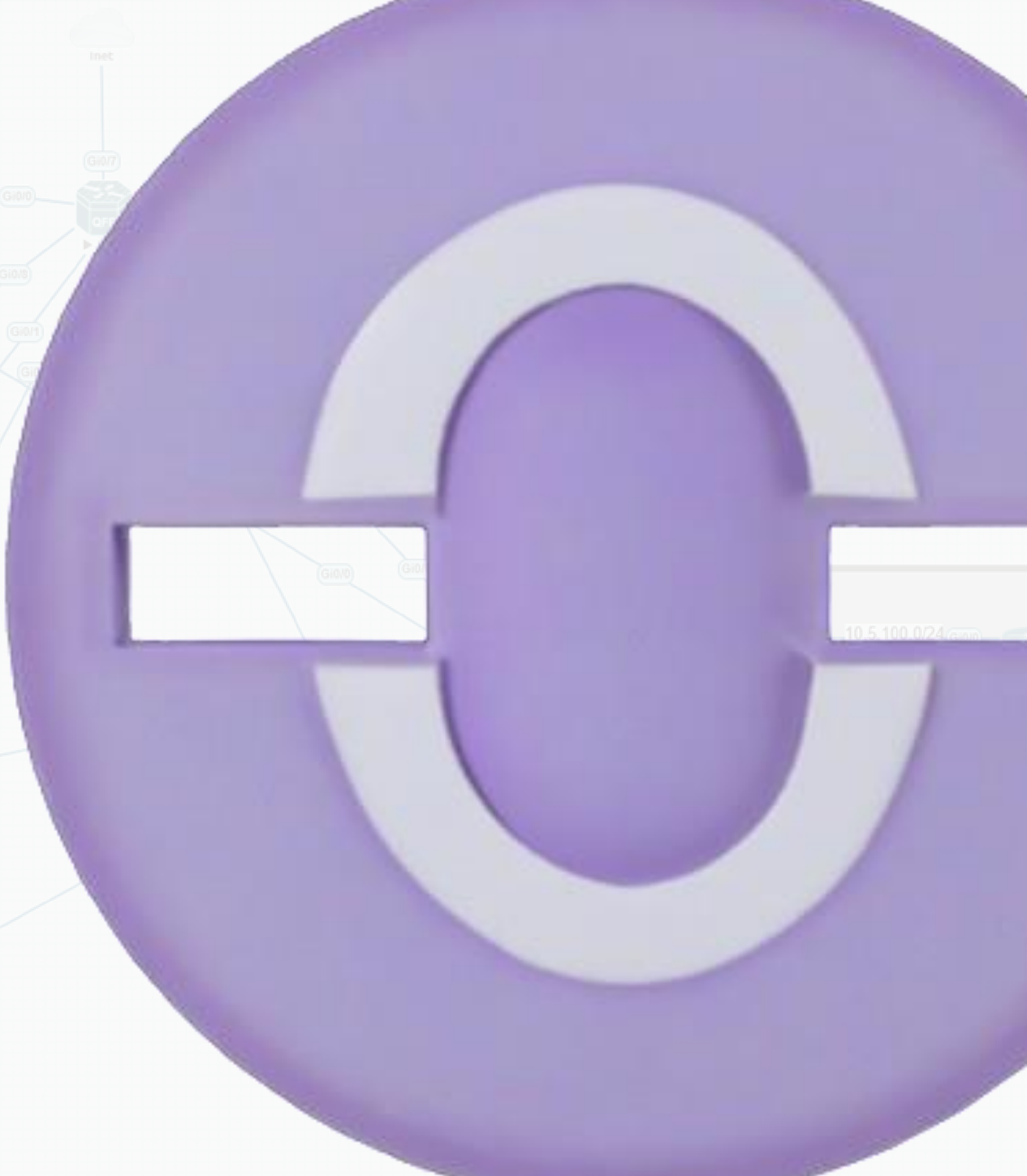
vManage (manager)

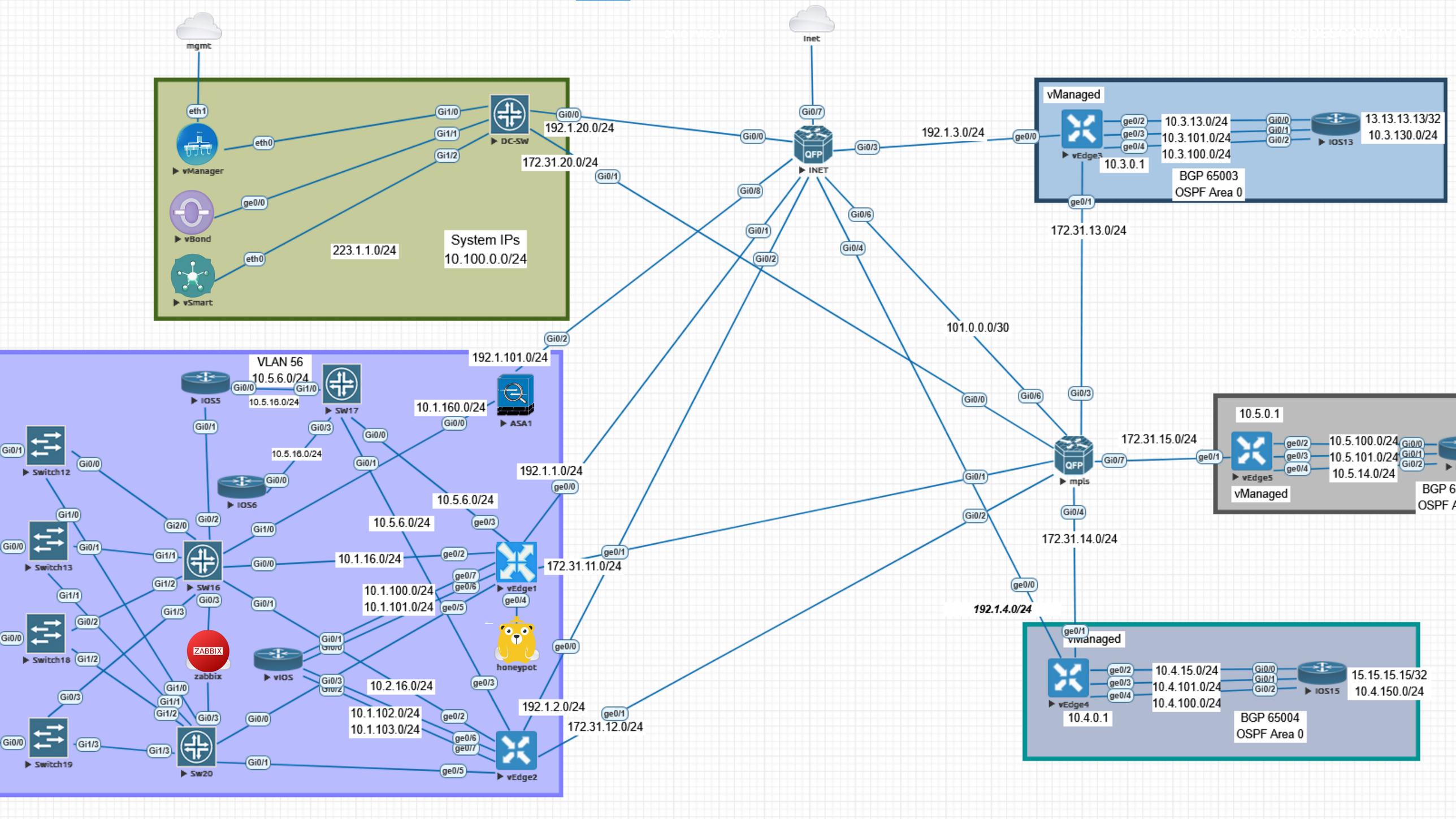
Definition: Centralized network management platform for Cisco SD-WAN, enabling configuration, monitoring,
Usage: Automates network operations, enforces security policies, and provides visibility across distributed SD-WAN environments


mgmt
Inet
eth1
eth0
vManager
vBond
vSmart
Gi1/0
Gi1/1
Gi1/2
DC-SW
Gi0/0
192.1.20.0/24
172.31.20.0/24
223.1.1.0/24
System IPs
10.100.0.0/24
Gi0/7
QFP
INET
Gi0/0
Gi0/1
Gi0/2
Gi0/3
Gi0/4
Gi0/6
Gi0/8
192.1.3.0/24
vManaged
ge0/0
ge0/1
ge0/2
ge0/3
ge0/4
vEdge3
10.3.0.1
10.3.13.0/24
10.3.101.0/24
10.3.100.0/24
IOS13
13.13.13.13/32
10.3.130.0/24
BGP 65003
OSPF Area 0
172.31.13.0/24
101.0.0.0/30
VLAN 56
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IOS5
SW17
ASA1
10.5.16.0/24
10.1.160.0/24
192.1.101.0/24
192.1.1.0/24
IOS6
Switch12
Switch13
Switch18
Switch19
SW16
Sw20
Gi2/0
10.5.6.0/24
10.1.16.0/24
10.1.100.0/24
10.1.101.0/24
ZABBIX
zabbix
vIOS
honeypot
10.2.16.0/24
10.1.102.0/24
10.1.103.0/24
vEdge1
vEdge2
172.31.11.0/24
192.1.2.0/24
172.31.12.0/24
mpls
172.31.15.0/24
10.5.0.1
vEdge5
10.5.100.0/24
10.5.101.0/24
10.5.14.0/24
vManaged
BGP 6
OSPF A
172.31.14.0/24
192.1.4.0/24
vEdge4
10.4.0.1
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24
IOS15
15.15.15.15/32
10.4.150.0/24
BGP 65004
OSPF Area 0

vBond (validator)

Definition: Orchestration engine that authenticates and onboards SD-WAN devices (vEdges, vSmarts).
Usage: Securely registers new devices into the SD-WAN fabric and facilitates initial connectivity.



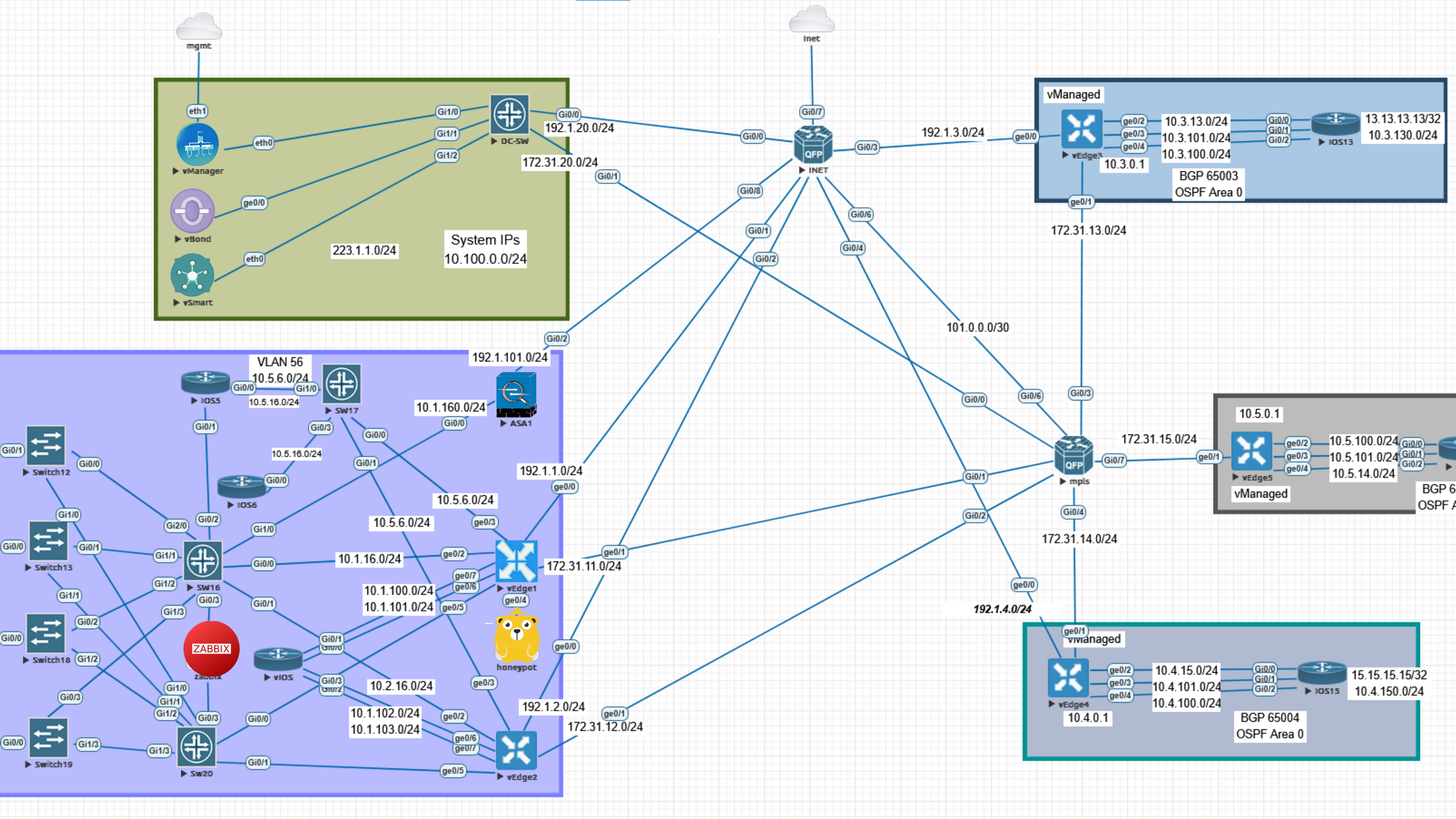

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10.5.16.0/24
IOS6
192.1.1.0/24
10.5.6.0/24
Switch12
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vEdge1
vEdge2
10.1.16.0/24
10.1.100.0/24
10.1.101.0/24
172.31.11.0/24
ZABBIX
zabbix
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10.2.16.0/24
10.1.102.0/24
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192.1.2.0/24
172.31.12.0/24
Gi0/6
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10.4.100.0/24
IOS15
15.15.15.15/32
10.4.150.0/24
BGP 65004
OSPF Area 0

vSmart (controller)

Definition: Controller responsible for traffic orchestration and policy distribution in Cisco SD-WAN.

Usage: Directs application traffic based on centralized policies (e.g., prioritizing VoIP) and ensures secure data routing.

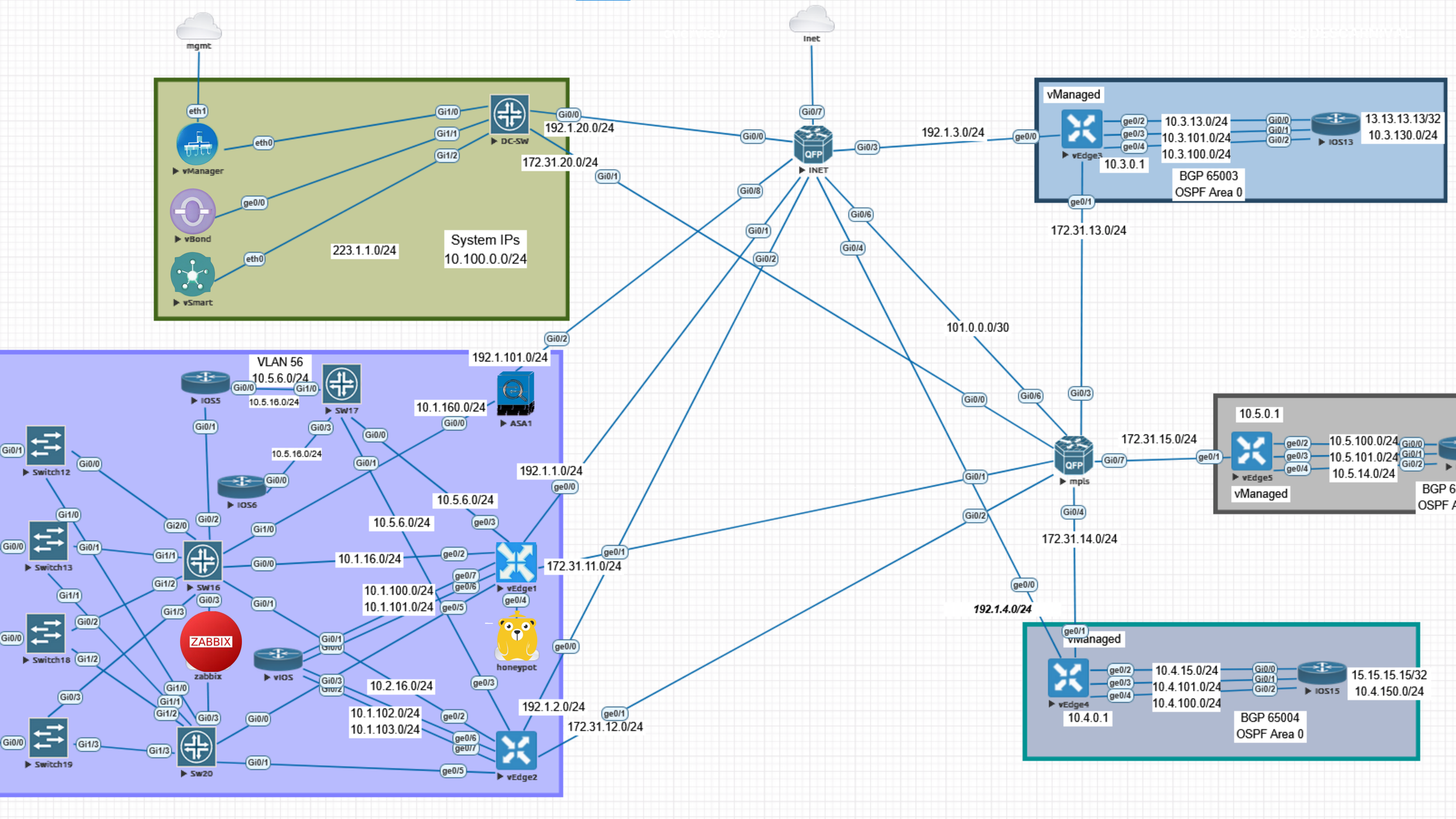



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ge0/1
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QFP
mpls
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10.5.0.1
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BGP 6
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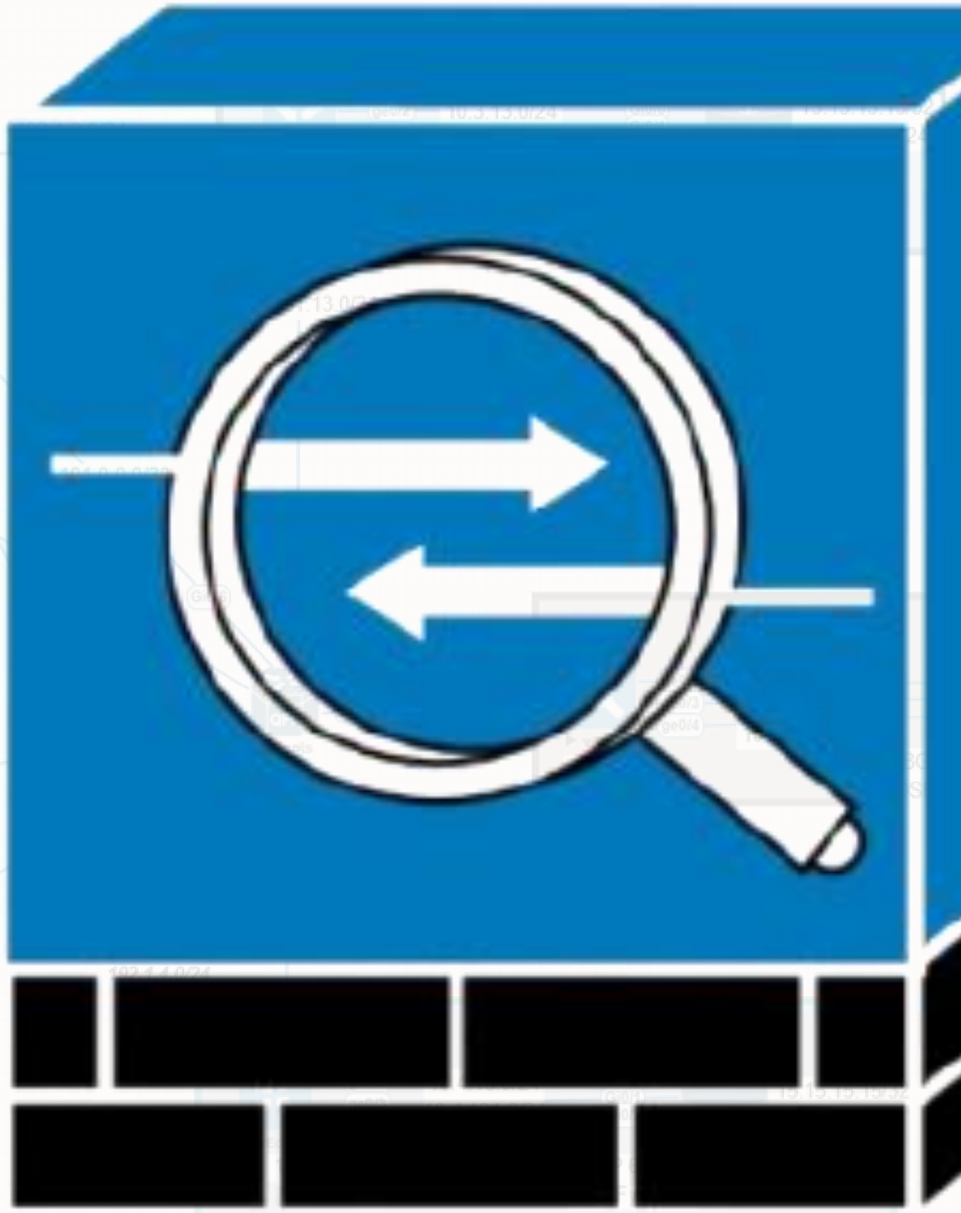
vEdge (edge)

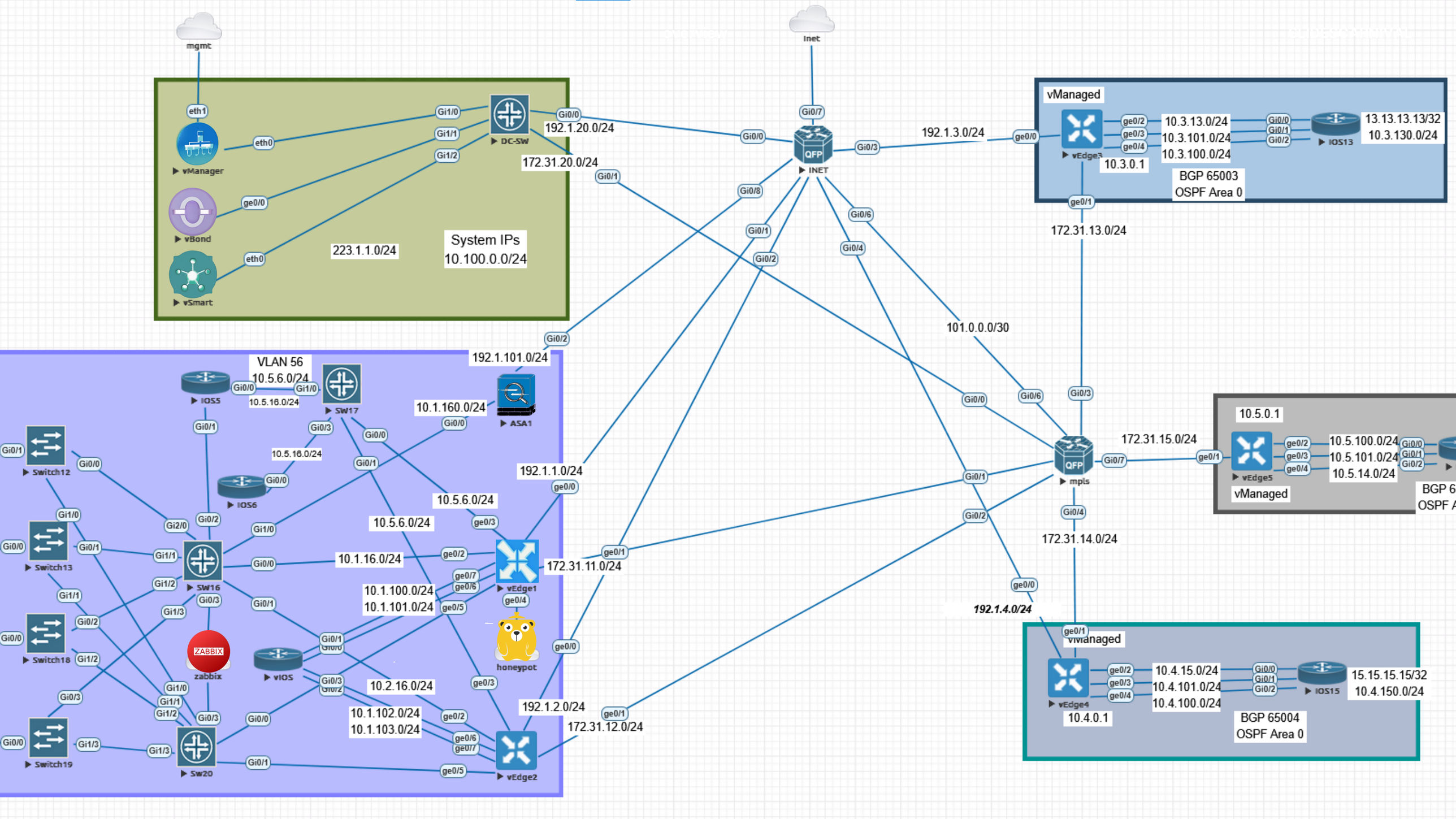
Definition:advanced Router deployed at branch/edge locations to connect sites to the SD-WAN overlay.
Usage: Provides secure WAN connectivity (e.g., linking branch offices to cloud services) and enforces local policies.


mgmt
Inet
eth1
eth0
vManager
vBond
vSmart
Gi1/0
Gi1/1
Gi1/2
DC-SW
Gi0/0
192.1.20.0/24
172.31.20.0/24
223.1.1.0/24
System IPs
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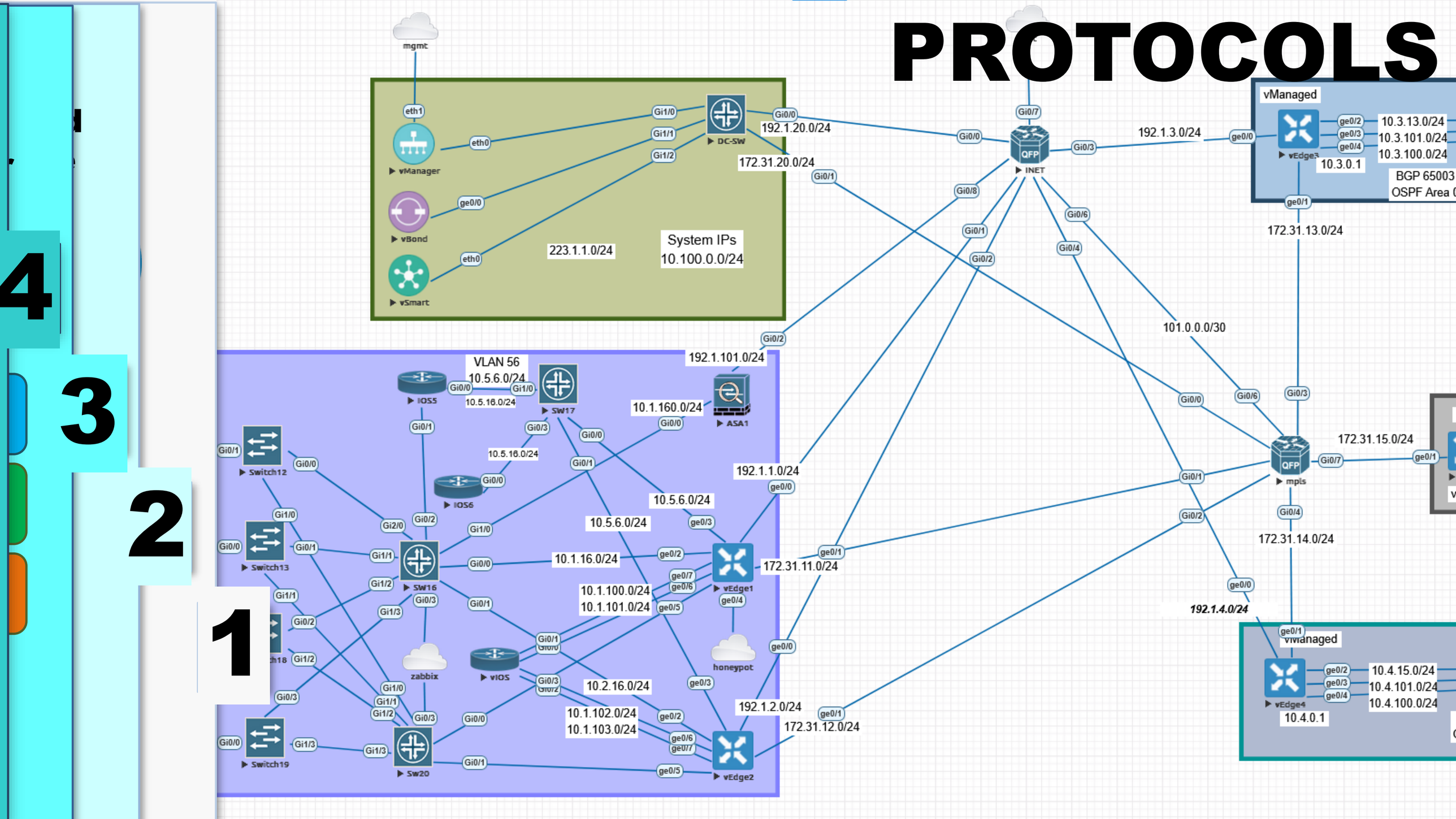
ASA

(Adaptive security appliance)
Definition: firewall appliance offering stateful inspection, VPN, and threat prevention.
Usage: Protects network perimeters (e.g., blocking unauthorized access) and secures remote connections via VPN




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PROTOCOLS


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172.31.14.0/24
ge0/0
192.1.4.0/24
ge0/1
vManaged
vEdge4
ge0/2
ge0/3
ge0/4
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24
10.4.0.1

ce)

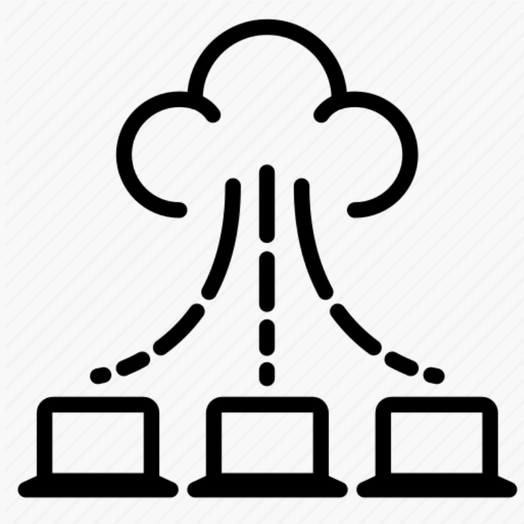
ing

4

3

2

- DHCP



- VRRP



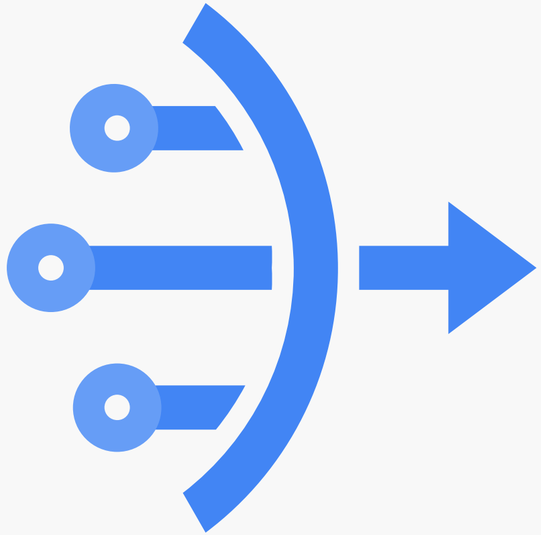
- HSRP



- BGP

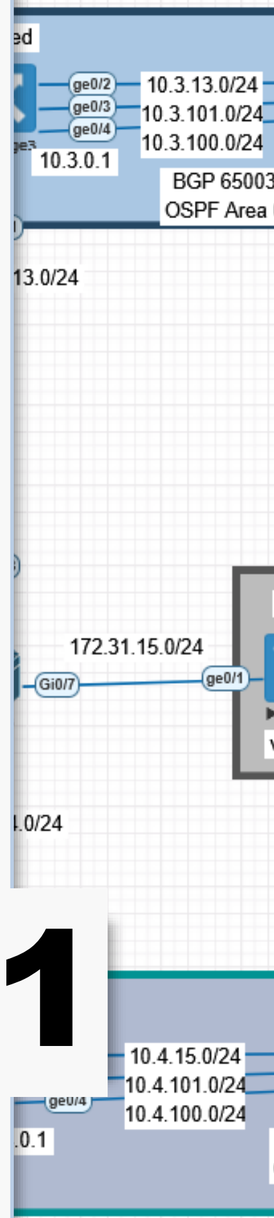


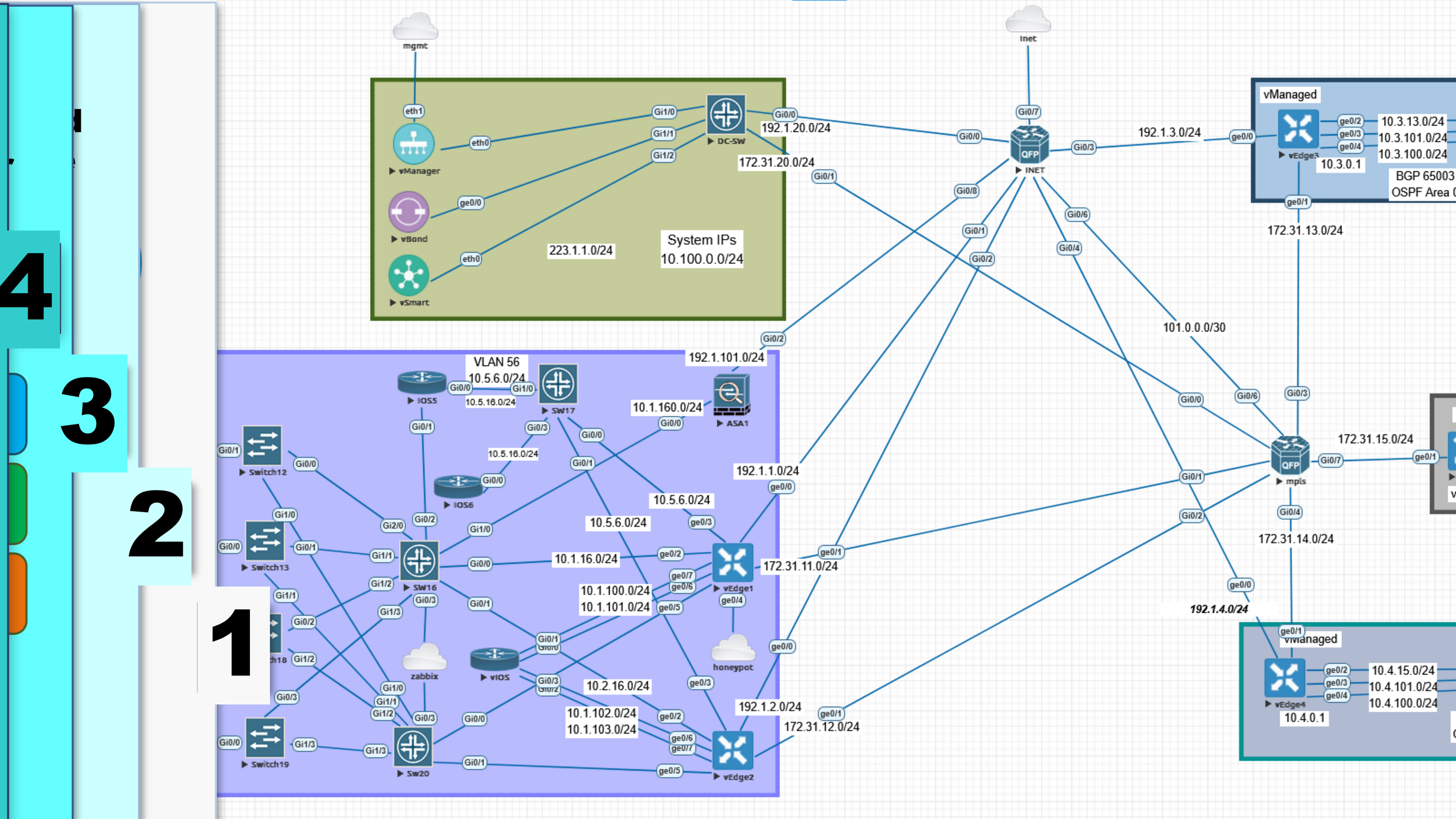
-NAT



- OSPF

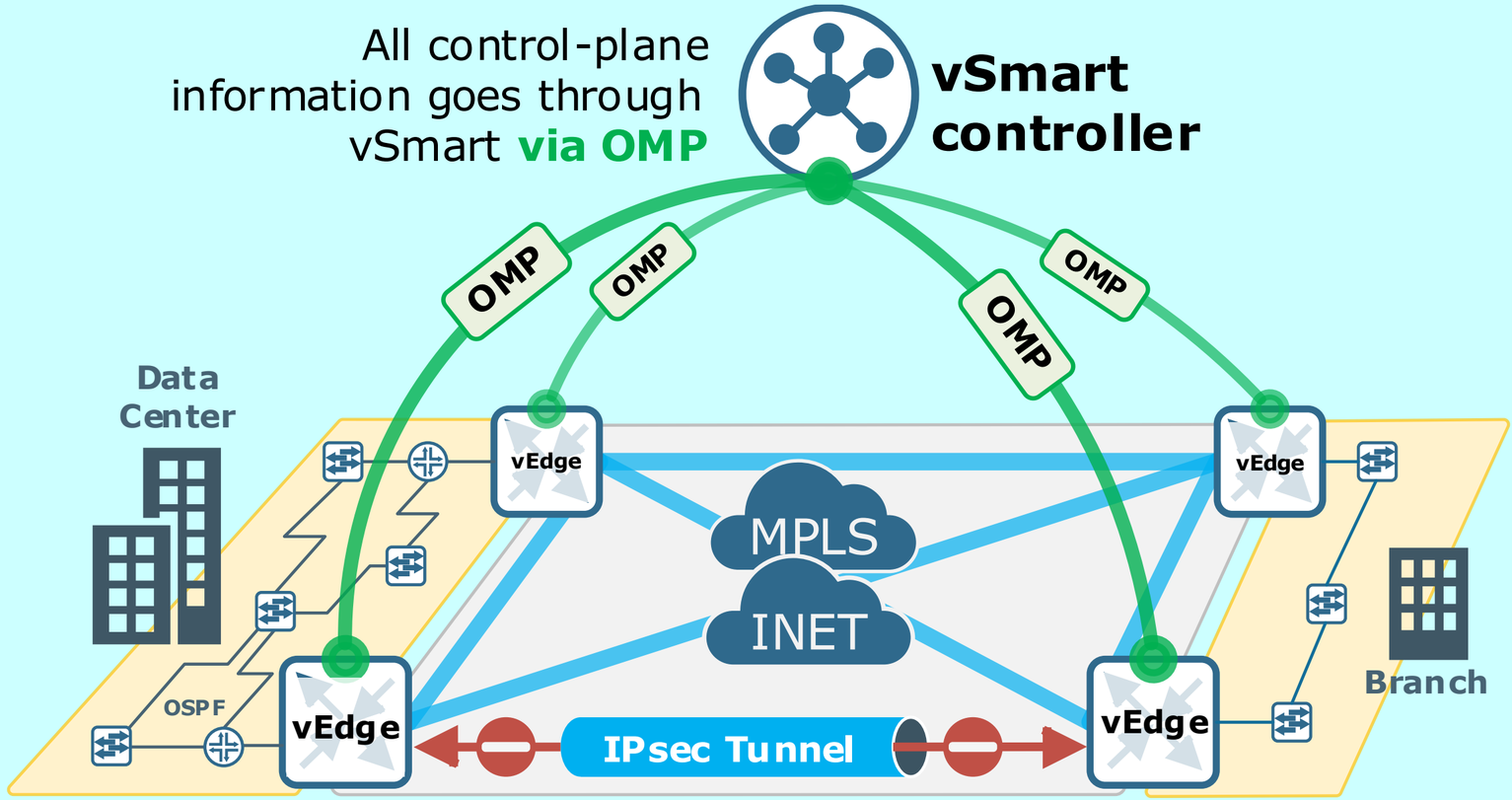




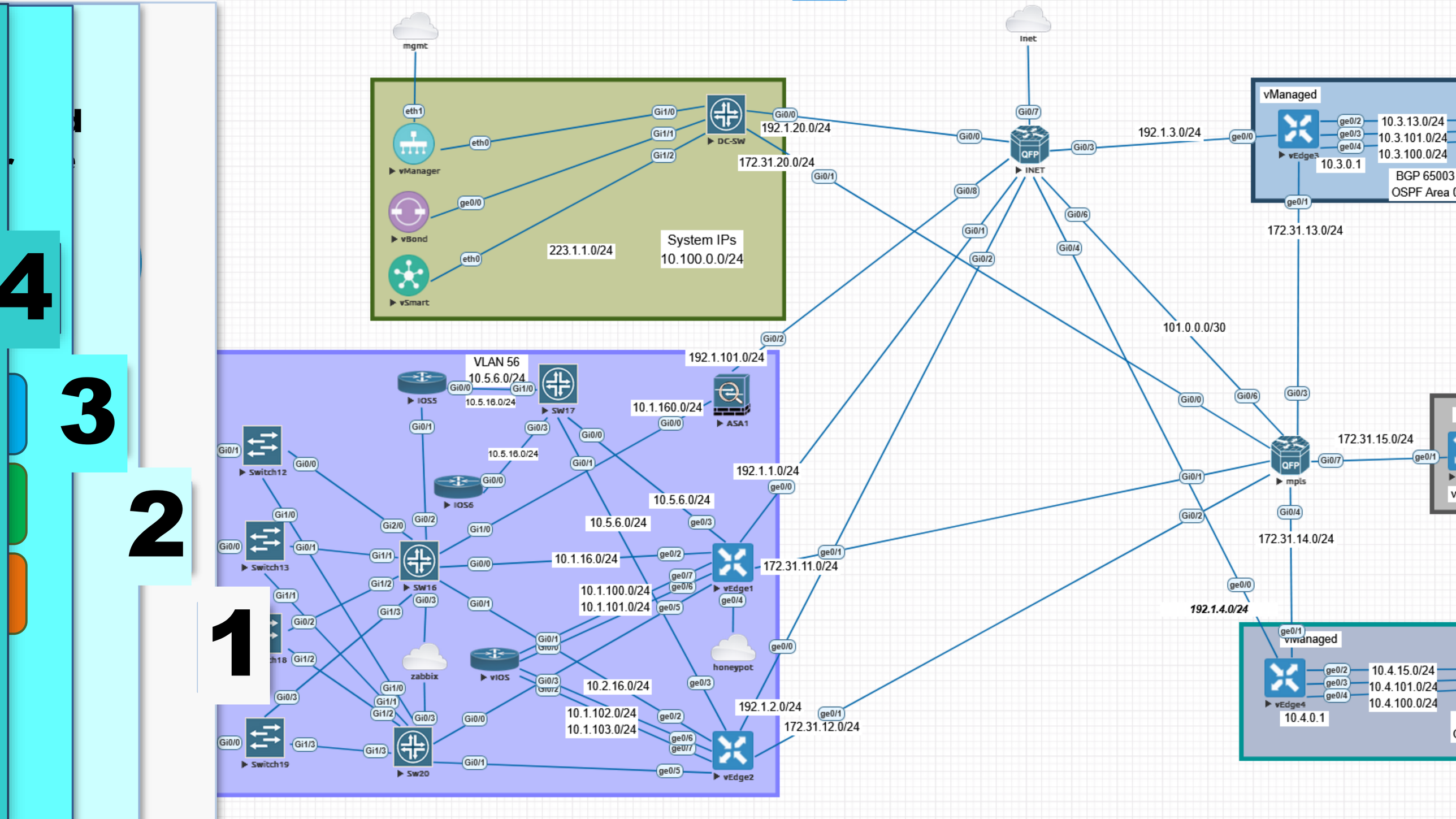

mgmt
eth1
vManager
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vBond
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vSmart
Gi1/0
Gi1/1
Gi1/2
DC-SW
Gi0/0
192.1.20.0/24
172.31.20.0/24
223.1.1.0/24
System IPs
10.100.0.0/24
Inet
Gi0/7
QFP
INET
Gi0/0
Gi0/3
192.1.3.0/24
vManaged
ge0/0
vEdge3
ge0/2
ge0/3
ge0/4
10.3.13.0/24
10.3.101.0/24
10.3.100.0/24
10.3.0.1
BGP 65003
OSPF Area
ge0/1
172.31.13.0/24
Gi0/1
Gi0/8
Gi0/6
Gi0/4
Gi0/2
101.0.0.0/30
Gi0/2
192.1.101.0/24
VLAN 56
10.5.6.0/24
IOS5
Gi0/0
Gi1/0
10.5.16.0/24
SW17
ASA1
10.1.160.0/24
Gi0/1
Gi0/3
Gi0/0
10.5.16.0/24
Switch12
Gi0/1
Gi0/0
IOS6
Gi0/0
Gi0/0
Gi0/6
Gi0/3
Gi0/0
Gi1/0
Gi0/0
Gi0/1
Gi2/0
Gi0/2
Gi1/0
10.5.6.0/24
10.5.6.0/24
ge0/3
192.1.1.0/24
ge0/0
Switch13
Gi0/0
Gi0/1
Gi1/1
SW16
Gi0/0
10.1.16.0/24
ge0/2
vEdge1
ge0/7
ge0/6
ge0/4
ge0/5
10.1.100.0/24
10.1.101.0/24
Gi1/2
Gi0/3
Gi1/3
Gi0/1
Gi1/1
Gi0/2
Gi0/1
zabbix
vIOS
honeypot
Gi1/2
Gi0/3
Gi1/0
Gi1/1
Gi1/2
Gi0/3
10.2.16.0/24
ge0/3
192.1.2.0/24
172.31.12.0/24
10.1.102.0/24
10.1.103.0/24
ge0/2
ge0/6
ge0/7
Gi0/0
Gi1/3
Gi1/3
Switch19
Sw20
Gi0/1
ge0/5
vEdge2
Gi0/1
Gi0/7
mpls
ge0/1
172.31.15.0/24
Gi0/1
Gi0/2
Gi0/4
172.31.14.0/24
172.31.11.0/24
ge0/0
192.1.4.0/24
ge0/1
vManaged
vEdge4
ge0/2
ge0/3
ge0/4
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24
10.4.0.1
ge0/1
172.31.12.0/24

- OMP (Overlay Management Protocol)

Cisco SD-WAN protocol that distributes routing, policy, and security information between vSmart controllers and edge

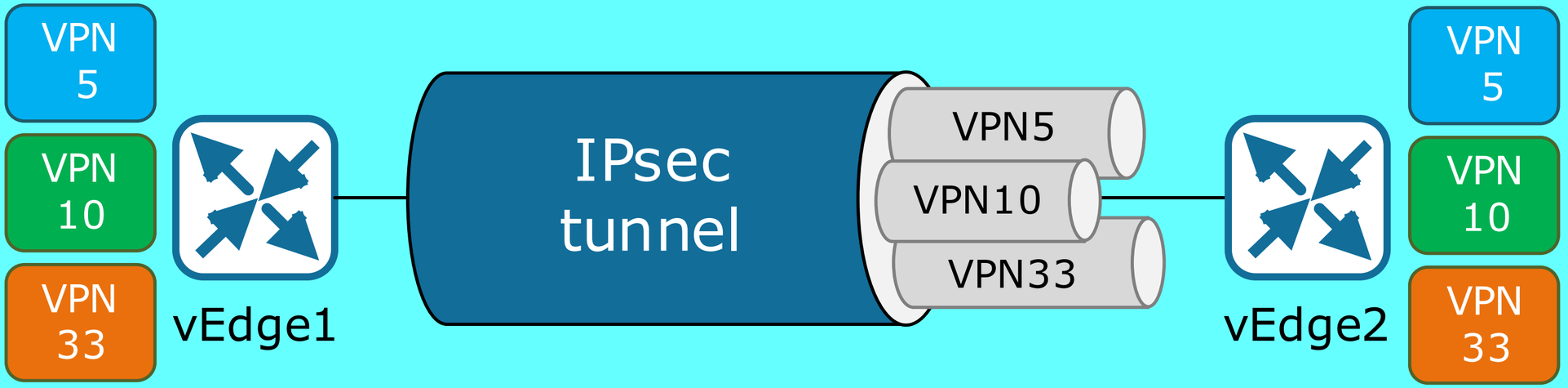


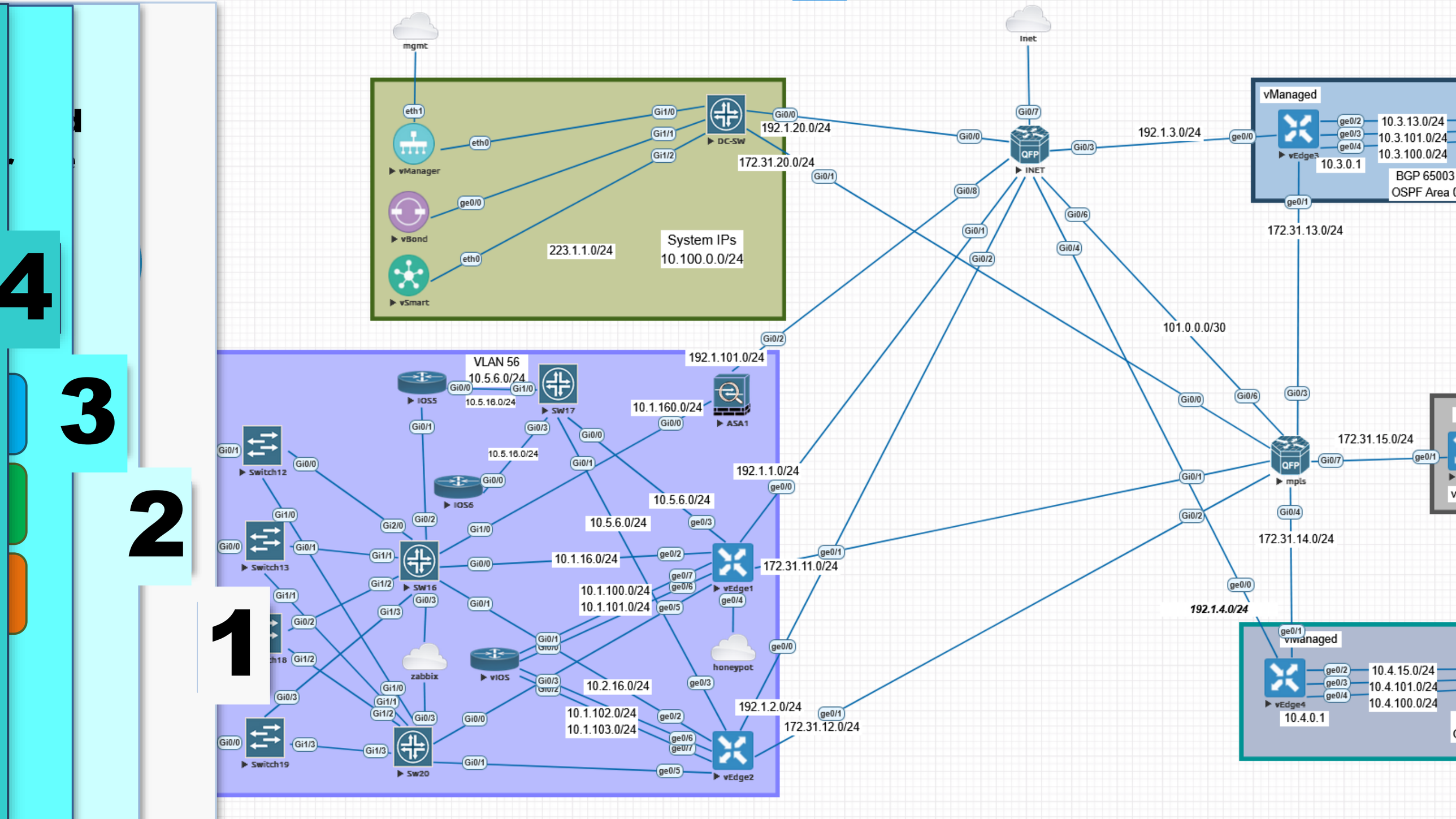
vEdges **never exchange** control plane information directly in-band through the data-plane


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223.1.1.0/24
System IPs
10.100.0.0/24
Inet
Gi0/7
QFP
INET
Gi0/0
Gi0/3
192.1.3.0/24
vManaged
ge0/0
vEdge3
ge0/2
ge0/3
ge0/4
10.3.13.0/24
10.3.101.0/24
10.3.100.0/24
10.3.0.1
BGP 65003
OSPF Area
ge0/1
172.31.13.0/24
Gi0/1
Gi0/8
Gi0/6
Gi0/4
Gi0/2
101.0.0.0/30
Gi0/2
192.1.101.0/24
VLAN 56
10.5.6.0/24
IOS5
Gi0/0
Gi1/0
10.5.16.0/24
SW17
ASA1
10.1.160.0/24
Gi0/1
Gi0/3
Gi0/0
10.5.16.0/24
Switch12
Gi0/1
Gi0/0
IOS6
Gi0/0
Gi0/0
Gi0/6
Gi0/3
Gi0/0
Gi1/0
Gi0/0
Gi0/1
Gi2/0
Gi0/2
Gi1/0
10.5.6.0/24
10.5.6.0/24
ge0/3
192.1.1.0/24
ge0/0
Switch13
Gi0/0
Gi0/1
Gi1/1
SW16
Gi0/0
10.1.16.0/24
ge0/2
vEdge1
ge0/7
ge0/6
ge0/4
ge0/5
10.1.100.0/24
10.1.101.0/24
Gi1/2
Gi0/3
Gi1/3
Gi0/1
Gi1/1
Gi0/2
Gi0/1
zabbix
vIOS
honeypot
Gi1/2
Gi0/3
Gi1/0
Gi1/1
Gi1/2
Gi0/3
10.2.16.0/24
ge0/3
192.1.2.0/24
172.31.12.0/24
10.1.102.0/24
10.1.103.0/24
ge0/2
ge0/6
ge0/7
Gi0/0
Gi1/3
Gi1/3
Switch19
Sw20
Gi0/1
ge0/5
vEdge2
Gi0/1
Gi0/7
mpls
ge0/1
172.31.15.0/24
Gi0/1
Gi0/2
Gi0/4
172.31.14.0/24
172.31.11.0/24
ge0/0
192.1.4.0/24
ge0/1
vManaged
vEdge4
ge0/2
ge0/3
ge0/4
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24
10.4.0.1
ge0/1
172.31.12.0/24

- VPN(virtual private network)

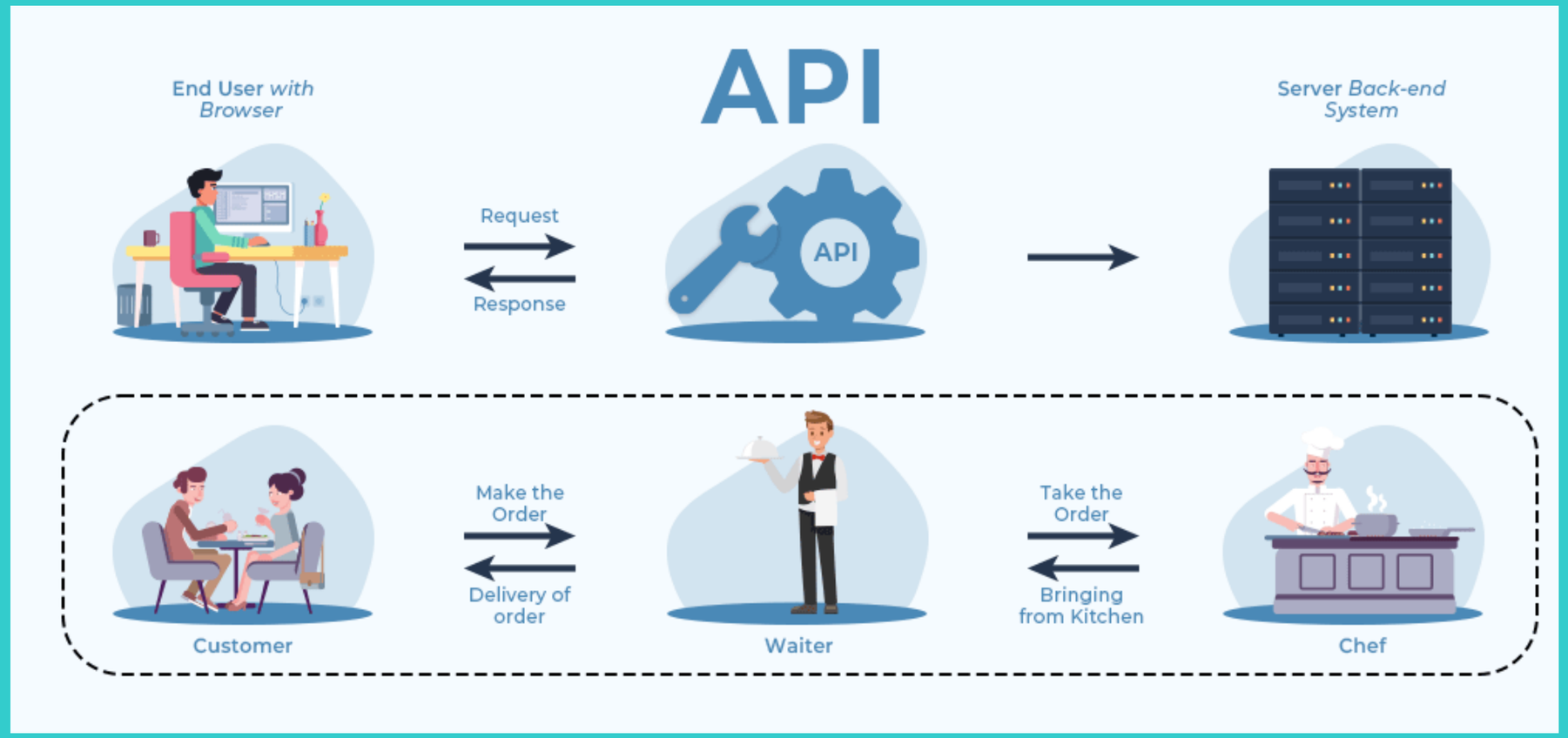
Isolates traffic into separate virtual networks (e.g., guest vs. corporate traffic) using encryption/VRFs for security and QoS policies

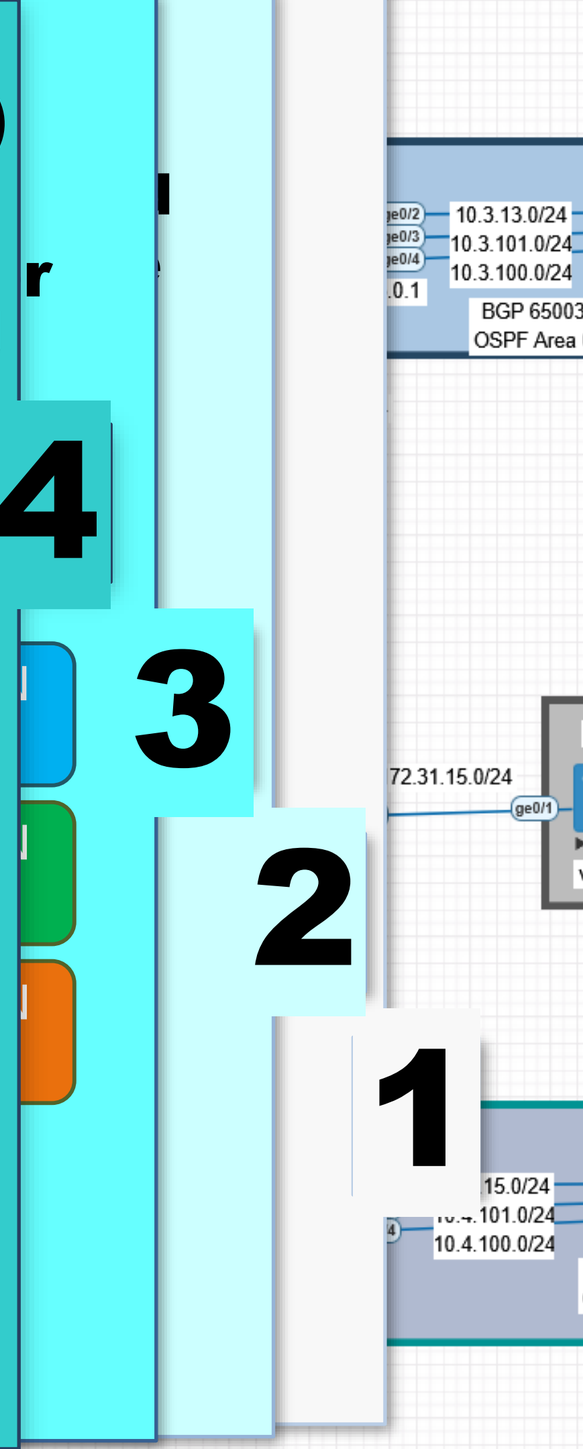



mgmt
eth1
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vBond
eth0
vSmart
Gi1/0
Gi1/1
Gi1/2
DC-SW
Gi0/0
192.1.20.0/24
172.31.20.0/24
223.1.1.0/24
System IPs
10.100.0.0/24
Inet
Gi0/7
QFP
INET
Gi0/0
Gi0/3
192.1.3.0/24
vManaged
ge0/0
vEdge3
ge0/2
ge0/3
ge0/4
10.3.13.0/24
10.3.101.0/24
10.3.100.0/24
10.3.0.1
BGP 65003
OSPF Area
ge0/1
172.31.13.0/24
Gi0/1
Gi0/8
Gi0/6
Gi0/4
Gi0/2
101.0.0.0/30
Gi0/2
192.1.101.0/24
VLAN 56
10.5.6.0/24
IOS5
Gi0/0
Gi1/0
10.5.16.0/24
SW17
ASA1
10.1.160.0/24
Gi0/1
Gi0/3
Gi0/0
10.5.16.0/24
Switch12
Gi0/1
Gi0/0
IOS6
Gi0/0
Gi0/0
Gi0/6
Gi0/3
Gi0/0
Gi1/0
Gi0/0
Gi0/1
Gi2/0
Gi0/2
Gi1/0
10.5.6.0/24
10.5.6.0/24
ge0/3
192.1.1.0/24
ge0/0
Switch13
Gi0/0
Gi0/1
Gi1/1
SW16
Gi0/0
10.1.16.0/24
ge0/2
vEdge1
ge0/7
ge0/6
ge0/4
ge0/5
10.1.100.0/24
10.1.101.0/24
Gi1/2
Gi0/3
Gi1/3
Gi0/1
Gi1/1
Gi0/2
Gi0/1
zabbix
vIOS
honeypot
Gi1/2
Gi0/3
Gi1/0
Gi1/1
Gi1/2
Gi0/3
10.2.16.0/24
ge0/3
192.1.2.0/24
172.31.12.0/24
10.1.102.0/24
10.1.103.0/24
ge0/2
ge0/6
ge0/7
Gi0/0
Gi1/3
Gi1/3
Switch19
Sw20
Gi0/1
ge0/5
vEdge2
Gi0/1
Gi0/7
mpls
ge0/1
172.31.15.0/24
Gi0/1
Gi0/2
Gi0/4
172.31.14.0/24
172.31.11.0/24
ge0/0
192.1.4.0/24
ge0/1
vManaged
vEdge4
ge0/2
ge0/3
ge0/4
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24
10.4.0.1
ge0/1
172.31.12.0/24

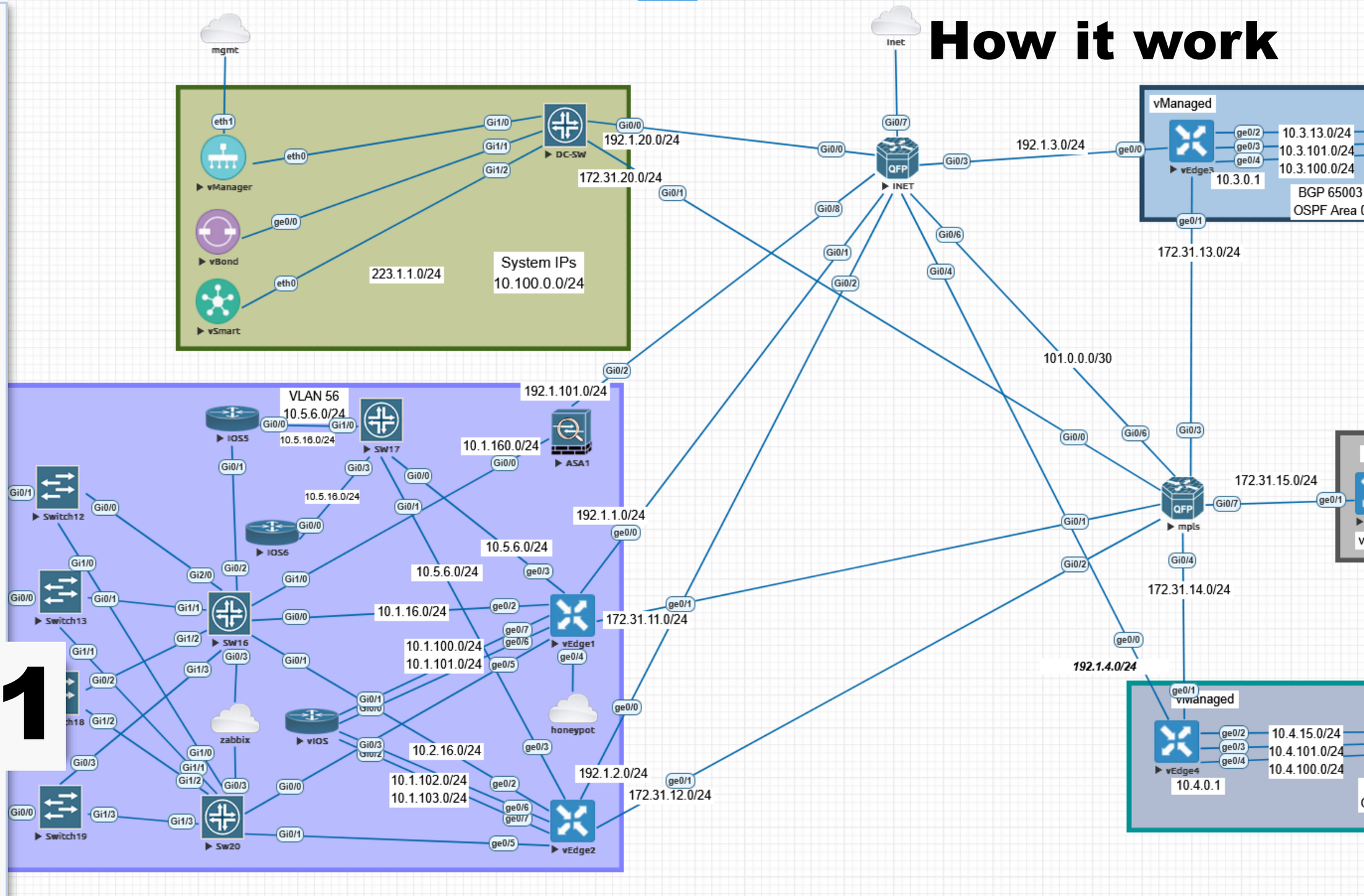
- API (Application Programming Interface)

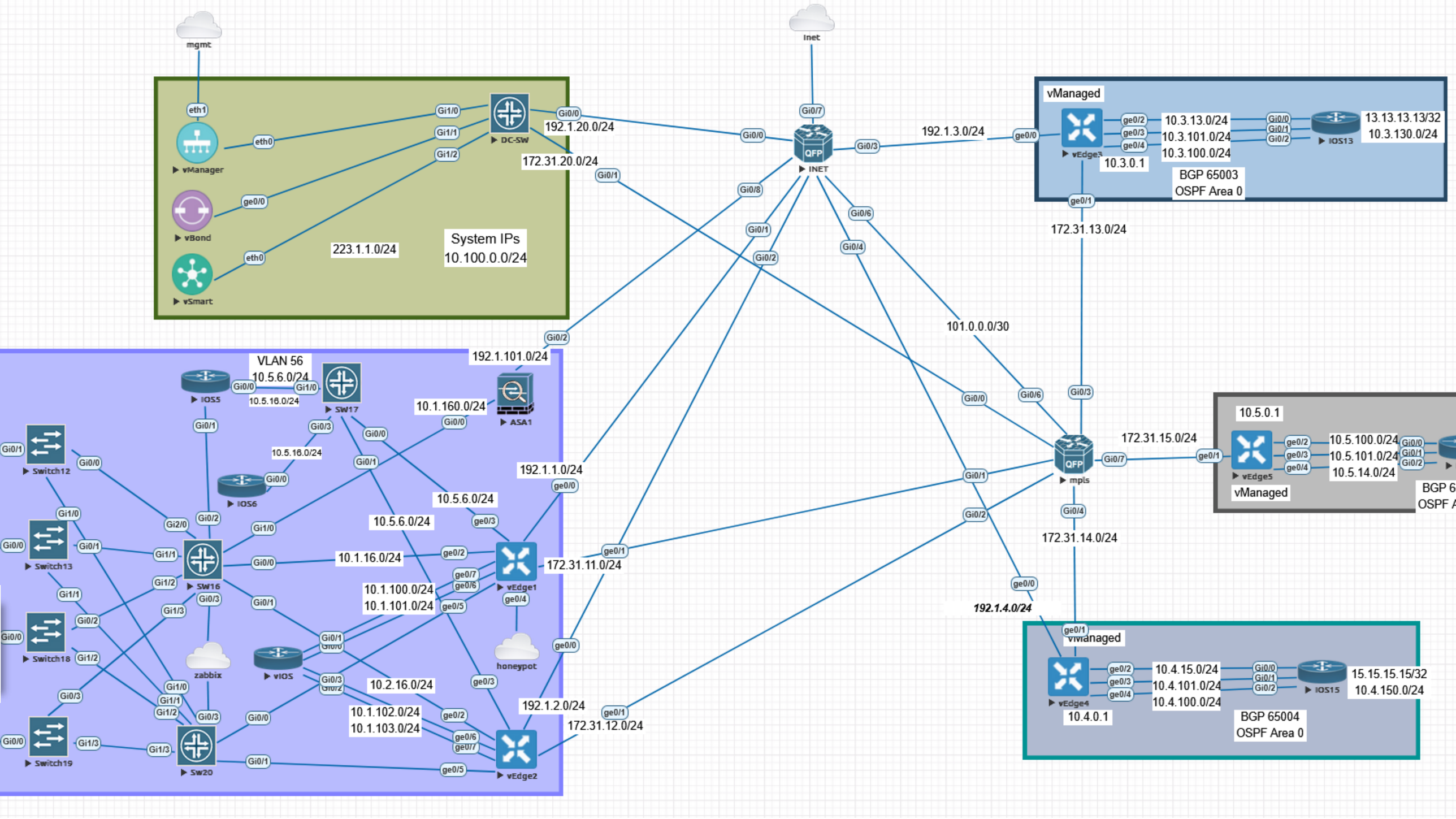
Enables automated control of SD-WAN components (like routers or policies) by allowing systems/software to communicate programmatically.





How it work


mgmt
eth1
vManager
eth0
vBond
eth0
vSmart
Gi1/0
Gi1/1
Gi1/2
DC-SW
Gi0/0
192.1.20.0/24
172.31.20.0/24
223.1.1.0/24
System IPs
10.100.0.0/24
Inet
Gi0/7
QFP
INET
Gi0/0
Gi0/1
Gi0/2
Gi0/3
Gi0/4
Gi0/6
Gi0/8
192.1.3.0/24
vManaged
ge0/0
ge0/1
ge0/2
ge0/3
ge0/4
vEdge3
10.3.0.1
10.3.13.0/24
10.3.101.0/24
10.3.100.0/24
BGP 65003
OSPF Area
172.31.13.0/24
101.0.0.0/30
VLAN 56
10.5.6.0/24
10.5.16.0/24
IOS5
SW17
ASA1
192.1.101.0/24
10.1.160.0/24
IOS6
Switch12
Switch13
Switch18
Switch19
SW16
Sw20
zabbix
vIOS
honeypot
vEdge1
vEdge2
10.1.16.0/24
10.1.100.0/24
10.1.101.0/24
10.2.16.0/24
10.1.102.0/24
10.1.103.0/24
192.1.1.0/24
192.1.2.0/24
172.31.11.0/24
172.31.12.0/24
mpls
172.31.15.0/24
172.31.14.0/24
192.1.4.0/24
vManaged
vEdge4
10.4.0.1
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24


mgmt
Inet
eth1
eth0
vManager
vBond
vSmart
DC-SW
Gi1/0
Gi1/1
Gi1/2
Gi0/0
192.1.20.0/24
172.31.20.0/24
223.1.1.0/24
System IPs
10.100.0.0/24
INET
QFP
Gi0/7
Gi0/1
Gi0/2
Gi0/3
Gi0/4
Gi0/6
Gi0/8
192.1.3.0/24
vManaged
vEdge3
ge0/0
ge0/1
ge0/2
ge0/3
ge0/4
10.3.13.0/24
10.3.101.0/24
10.3.100.0/24
10.3.0.1
IOS13
13.13.13.13/32
10.3.130.0/24
BGP 65003
OSPF Area 0
172.31.13.0/24
101.0.0.0/30
VLAN 56
10.5.6.0/24
IOS5
SW17
ASA1
192.1.101.0/24
10.1.160.0/24
10.5.16.0/24
IOS6
Switch12
Switch13
Switch18
Switch19
SW16
Sw20
192.1.1.0/24
10.5.6.0/24
10.1.16.0/24
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vEdge1
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10.1.102.0/24
10.1.103.0/24
192.1.2.0/24
172.31.12.0/24
172.31.11.0/24
mpls
172.31.15.0/24
10.5.0.1
vEdge5
10.5.100.0/24
10.5.101.0/24
10.5.14.0/24
vManaged
172.31.14.0/24
192.1.4.0/24
vEdge4
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24
10.4.0.1
IOS15
15.15.15.15/32
10.4.150.0/24
BGP 65004
OSPF Area 0

VPN	PREFIX	PROTOCOL	PROTOCOL SUB TYPE	NEXTHOP IF NAME	NEXTHOP ADDR	STATUS
0	0.0.0.0/0	static	---	ge0/0	192.1.1.1	F,S
0	10.12.0.1/32	connected	---	system	----	F,S
0	172.31.11.0/24	connected	---	ge0/1	----	F,S
0	172.31.13.0/24	bgp	e	ge0/1	172.31.11.1	F,S
0	172.31.14.0/24	bgp	e	ge0/1	172.31.11.1	F,S
0	172.31.15.0/24	bgp	e	ge0/1	172.31.11.1	F,S
0	192.1.1.0/24	connected	---	ge0/0	----	F,S
0	223.1.1.0/24	bgp	e	ge0/1	172.31.11.1	F,S

```
vedge1# show run sys
system
 host-name                 vedge1
 system-ip                 10.12.0.1
 site-id                   12
 admin-tech-on-failure
 no route-consistency-check
 organization-name         lab
 vbond 223.1.1.11
 aaa
  auth-order local radius tacacs
  usergroup basic
   task system read write
   task interface read write
  !
  usergroup netadmin
  !
  usergroup operator
   task system read
   task interface read
   task policy read
   task routing read
   task security read
  !
  usergroup tenantadmin
  !
  user admin
   password $6$S4Vb1lbZu.2vj8FP$fkzMPSApdvixsO/cl1fwp67c2JryBNABu9nuy3AxULOZ6QLembHQ045pbGKbSKUVnVRV0OsrZU3WDnE
  !
 !
 logging
  disk
   enable
  !
 !
!
vedge1#
```

```
vedge1# show running-config vpn 0
vpn 0
 router
  bgp 65012
   address-family ipv4-unicast
    network 172.31.11.0/24
   !
   neighbor 172.31.11.1
    no shutdown
    remote-as 100
    address-family ipv4-unicast
    !
   !
  !
 !
 interface ge0/0
  ip address 192.1.1.2/24
  ipv6 dhcp-client
  tunnel-interface
   encapsulation ipsec
   color public-internet restrict
   allow-service all
   no allow-service bgp
   allow-service dhcp
   allow-service dns
   allow-service icmp
   no allow-service sshd
   no allow-service netconf
   no allow-service ntp
   no allow-service ospf
   no allow-service stun
   allow-service https
  !
  no shutdown
 !
 interface ge0/1
  ip address 172.31.11.2/24
  tunnel-interface
   encapsulation ipsec
   color mpls restrict
   allow-service all
   no allow-service bgp
   allow-service dhcp
   allow-service dns
   allow-service icmp
   no allow-service sshd
   no allow-service netconf
   no allow-service ntp
   no allow-service ospf
   no allow-service stun
   allow-service https
  !
  no shutdown
 !
 ip route 0.0.0.0/0 192.1.1.1
!
```

vEdge1

```
vedge1# show bfd sessions
                                                
```

```
-----BEGIN CERTIFICATE-----
MIIDmjCCAoICCQDeQiGoyAXFHjANBgkqhkiG9w0BAQsFADBbMQswCQYDVQQGEwJV
SzESMBAGA1UECAwJSGFtcHNoaXJlMRQwEgYDVQQHDAtTb3V0aGFtcHRvbjEMMAoG
A1UECgwDbGFiMRQwEgYDVQQDDAtyb2dlci5sb2NhbDAeFw0yNDExMTIwNjAxNTha
Fw0zMDA1MDUwNjAxNThaMIHCMQswCQYDVQQGEwJVUzETMBEGA1UECBMKQ2FsaWZv
cm5pYTERMA8GA1UEBxMIU2FuIEpvc2UxDDAKBgNVBAsTA2xhYjEUMBIGA1UEChML
dklQdGVsYSBJbmMxQzBBBgNVBAMTOnZtYW5hZ2UtOWI1NzZhNTktY2ZhZi00ZTRl
LTgxYjMtMTAzZmE0MmE0Yzk2LTAudmlwdGVsYS5jb20xIjAgBgkqhkiG9w0BCQEW
E3N1cHBvcnRAdmlwdGVsYS5jb20wggEiMA0GCSqGSIb3DQEBAQUAA4IBDwAwggEK
AoIBAQCUpVmNjQmuR3SbmF7R2zlcrg8hmnxnVwysF8pWoElnyQjvtm7mN42PLirm
NE/ecaTPVGLP7Edd1Bly6Z4M/svZhzYfaSojO4rBOoDGwHh3iiKivuCmVDHJHmtT
Q2//mjliQ9EtohgkNxiynvcvcYeynLU3p9lJ6AOPfOlWW/9/G+oxsBu+/O6mccV6
IM87/FAUdve408hww5RmGJevdCQF13JzP+4sVwc95h2i1/p1s3gt/ZTsuzbp3B7m
KyVYflTtzimWkHvnUG9cZuZ+XGxdOgfWXBWpJ8gVsY/9UIhIVFyU+R5N+vd9YJii
ud8ggJ6vYD51tw5cD1PDQ8dN1QQTAgMBAAEwDQYJKoZIhvcNAQELBQADggEBAIcU
4/vpFC2S/2o+16IbKGwlIngb6am53vamRTOlU8KXz+FSctFCHEIC7VabMeUfJwV5
9RUWSzJnDgdKF/jxMGWwKSkX+lYLp4K+Os8GvR75uKl7T4vGS58Zlo2nasOE+NVF
mzhGowPR6S/hYctlYWyPJ+EDyAEFau07eCjmskWf1X/V1ur6ousW8rDB1S2UiDtW
P5GfEKJHKKm/bH7cu/fnGBeX1vW2/RzcbSEmdiZaBW5O5+EDfVrCf+nayiUIajdU
Ho6/EK1ypfwz38LFBTMM8On1LAW3hOSV/LZjvPwTQUBnly6o5yqaZ+YgZi/MgbAE
QiLUHPzE2DSgYOOfRXM=
-----END CERTIFICATE-----
~
"vmanage.crt" 22 lines, 1310 characters
```

DASHBOARD

vSmart	WAN Edge	vBond	vManage	Reboot	Warning	Invalid
1 ↑	5 ↑	1 ↑	1 ✓	0 (Last 24 hrs)	0	0
vSmart - 1	WAN Edge - 5	vBond - 1	vManage - 1			

Control Status (Total 6)

Control Up	6
Partial	0
Control Down	0

Site Health (Total 4)

Full WAN Connectivity	4 sites
Partial WAN Connectivity	0 sites
No WAN Connectivity	0 sites

Transport Interface Distribution

< 10 Mbps	0
10 Mbps - 100 Mbps	0
100 Mbps - 500 Mbps	0
> 500 Mbps	0

View Percent Utilization

WAN Edge Inventory

Total	5
Authorized	5
Deployed	5
Staging	0

WAN Edge Health (Total 5)

Normal	Warning	Error
5	0	0

Transport Health

Type: By Loss



Top Applications

No data to display

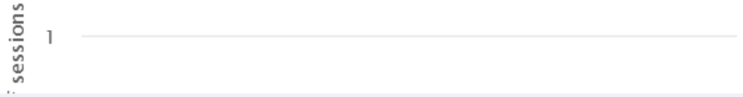
Application-Aware Routing

Type: By Loss

Tunnel Endpoints	Avg. Latency (ms)	Avg. Loss (%)	Avg. Jitter (ms)
vEdge4:mpls-vEdge3:mpls	48.417	0.07	16.021
vEdge4:mpls-vEdge5:mpls	46.458	0.07	14.208
vEdge4:mpls-vEdge2:mpls	41.271	0.063	12.5

FireWall Enforcement

Inspected | Dropped



Top Signature Hits

By Severity | By Count

URL Filtering

Blocked | Allowed

CONFIGURATION | TEMPLATES

Device | Feature

Add Template

Template Type: Non-Default

Search Options

Total Rows: 49

Name	Description	Type	Device Model	Device Templates	Devices Attached	Updated By	Last Updated
vEdge_Banner_template	vEdge_Banner_template	Banner	vEdge Cloud	3	4	admin	30 Nov 2024 9:37:06 ...
vEdge_system_template	vEdge_system_template	WAN Edge System	vEdge Cloud	3	4	admin	09 Dec 2024 2:41:50 ...
vEdge_VPN0_template	vEdge_VPN0_template	WAN Edge VPN	vEdge Cloud	1	2	admin	28 Jan 2025 4:58:06 ...
vEdge_Dual_Site_VPN101_template	vEdge_Dual_Site_VPN101_t...	WAN Edge VPN	vEdge Cloud	1	1	admin	31 Jan 2025 12:42:37...
vEdge_VPN512_template	vEdge_VPN512_template	WAN Edge VPN	vEdge Cloud	1	2	admin	09 Dec 2024 2:23:55 ...
vEdge_VPN0_int_G0/0_template	vEdge_VPN0_int_G0/0_tem...	WAN Edge Interface	vEdge Cloud	1	2	admin	28 Jan 2025 4:34:37 ...
vEdge_VPN512_int_Eth0	vEdge_VPN512_int_Eth0	WAN Edge Interface	vEdge Cloud	1	2	admin	09 Dec 2024 2:29:36 ...
vEdge_VPN0_int_G0/1_template	vEdge_VPN0_int_G0/1_tem...	WAN Edge Interface	vEdge Cloud	1	2	admin	09 Dec 2024 2:28:32 ...
vEdge_VPN0_BGP_Template	vEdge_VPN0_BGP_Template	BGP	vEdge Cloud	1	2	admin	09 Dec 2024 2:32:38 ...
vEdge_Single_Site_VPN1_Int_G0/2_template	vEdge_Single_Site_VPN1_I...	WAN Edge Interface	vEdge Cloud	1	2	admin	09 Dec 2024 6:36:41 ...
vEdge_Single_Site_VPN1_template	vEdge_Single_Site_VPN1_t...	WAN Edge VPN	vEdge Cloud	1	2	admin	28 Jan 2025 5:00:30 ...
vEdge_MPLS_Only_VPN512_int_Eth0	vEdge_MPLS_Only_VPN51...	WAN Edge Interface	vEdge Cloud	1	1	admin	09 Dec 2024 4:13:47 ...
vEdge_Dual_Site_VPN0_int_G0/0_template	vEdge_Dual_Site_VPN0_int...	WAN Edge Interface	vEdge Cloud	1	1	admin	09 Dec 2024 4:01:38 ...
vEdge_MPLS_Only_VPN0_template	vEdge_MPLS_Only_VPN0_t...	WAN Edge VPN	vEdge Cloud	1	1	admin	09 Dec 2024 4:29:48 ...
vEdge_MPLS_Only_VPN512_template	vEdge_MPLS_Only_VPN51...	WAN Edge VPN	vEdge Cloud	1	1	admin	09 Dec 2024 4:13:13 ...
vEdge_Dual_Site_bgp_vpn1_template	vEdge_Dual_Site_bgp_vpn1...	BGP	vEdge Cloud	1	1	admin	11 Dec 2024 2:54:20 ...
vEdge_Dual_Site_VPN512_int_Eth0	vEdge_Dual_Site_VPN512_i...	WAN Edge Interface	vEdge Cloud	1	1	admin	09 Dec 2024 4:02:17 ...
vEdge_Dual_Site_VPN0_template	vEdge_Dual_Site_VPN0_te...	WAN Edge VPN	vEdge Cloud	1	1	admin	09 Dec 2024 4:00:24 ...
vEdge_Dual_Site_VPN512_template	vEdge_Dual_Site_VPN512_t...	WAN Edge VPN	vEdge Cloud	1	1	admin	09 Dec 2024 4:01:10 ...
vEdge_Dual_Site_VPN0_int_G0/1_template	vEdge_Dual_Site_VPN0_int...	WAN Edge Interface	vEdge Cloud	1	1	admin	09 Dec 2024 4:02:32 ...

Select Device
Applications
Interface
TCP Optimization
WAN Throughput
Flows
Top Talkers
WAN
TLOC
Tunnel
Security Monitoring
Firewall
Intrusion Prevention
URL Filtering
Umbrella DNS Re-direct
Control Connections
System Status
Events
ACL Logs

vEdge2 | 10.12.0.2 Site ID: 12 Device Model: vEdge Cloud

Reboot	9
Crash	13
Module	N/A
Temperature Sensors	N/A
USB	N/A
Power Supply	N/A
Fans	N/A

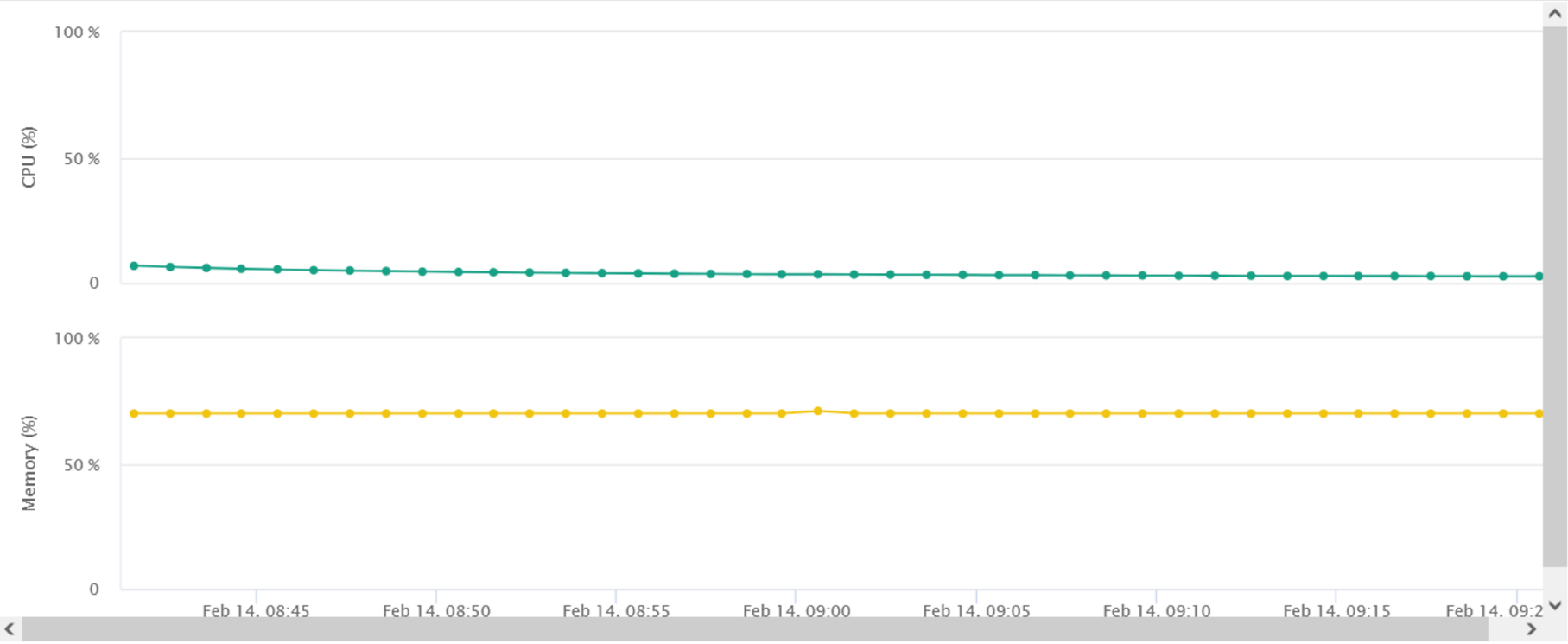
CPU & Memory

Real Time 1h 3h 6h 12h 24h 7days Custom

3.78%
CPU

Load average over 24 hrs

70.03%
Memory

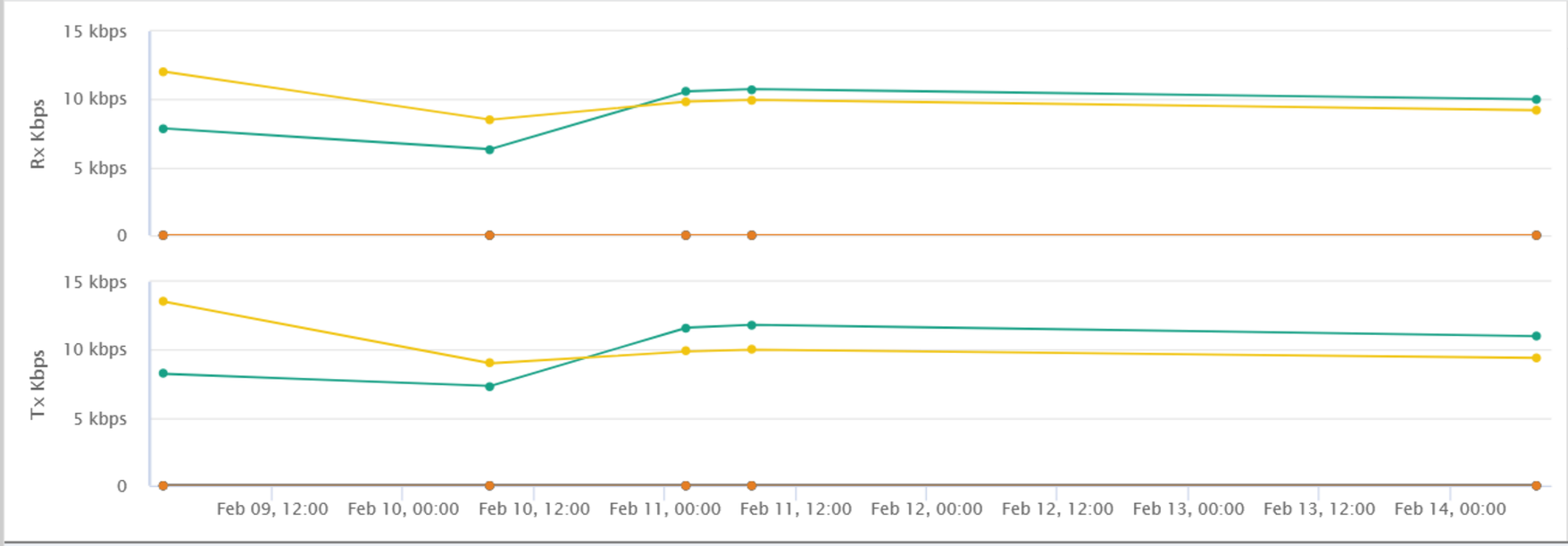


Select Device

vEdge2 | 10.12.0.2 Site ID: 12 Device Model: vEdge Cloud

Applications
Interface
TCP Optimization
WAN Throughput
Flows
Top Talkers
WAN
TLOC
Tunnel
Security Monitoring
Firewall
Intrusion Prevention
URL Filtering
Umbrella DNS Re-direct
Control Connections
System Status
Events
ACL Logs

Chart Options IPv4 & IPv6 Real Time 1h 3h 6h 12h 24h 7days Custom



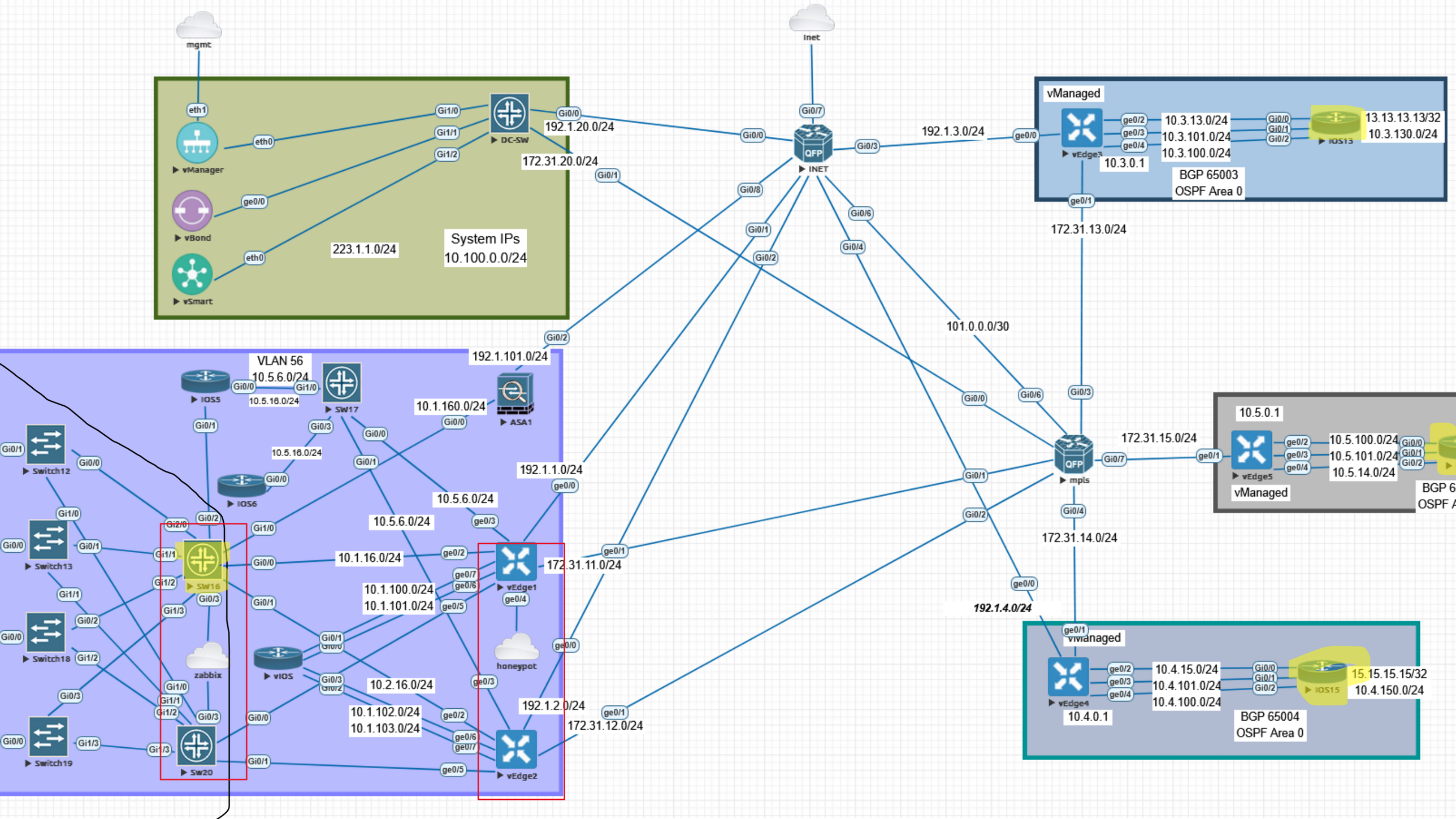
7 Rows Selected

Search Options

Total Rows: 12

Oper (4) Oper (8) Admin (4) Admin (8)

	Interface	Interface Description	IP Address	IPv6 Address	VPN	Admin Status	Oper Status	MTU	Speed (mbps)	TCP M
☑	ge0/0	INET_INTERFACE	192.1.2.2/24	-	0	↑	↑	1500	1000	1416
☑	ge0/1	MPLS_INTERFACE	172.31.12.2/24	-	0	↑	↑	1500	1000	1416
☑	system	--	10.12.0.2/32	-	0	↑	↑	1500	0	1416
☑	ge0/2	--	10.2.16.2/24	-	1	↑	↑	1500	1000	1416
☑	ge0/3	--	10.5.6.2/24	-	1	↑	↑	1500	1000	1416
☑	loopback0	--	10.1.0.2/32	-	1	↑	↑	1500	10	1416
☑	ge0/6	--	10.1.103.2/24	-	100	↑	↑	1500	1000	1416


mgmt
Inet
eth1
eth0
vManager
vBond
vSmart
Gi1/0
Gi1/1
Gi1/2
DC-SW
Gi0/0
192.1.20.0/24
172.31.20.0/24
223.1.1.0/24
System IPs
10.100.0.0/24
Gi0/7
QFP
INET
Gi0/1
Gi0/8
Gi0/6
Gi0/4
Gi0/2
Gi0/3
192.1.3.0/24
vManaged
ge0/0
ge0/2
ge0/3
ge0/4
vEdge3
10.3.0.1
10.3.13.0/24
10.3.101.0/24
10.3.100.0/24
IOS13
13.13.13.13/32
10.3.130.0/24
BGP 65003
OSPF Area 0
ge0/1
172.31.13.0/24
101.0.0.0/30
192.1.101.0/24
VLAN 56
10.5.6.0/24
IOS5
10.5.16.0/24
SW17
ASA1
10.1.160.0/24
Switch12
IOS6
192.1.1.0/24
10.5.6.0/24
Switch13
SW16
10.1.16.0/24
10.1.100.0/24
10.1.101.0/24
vEdge1
172.31.11.0/24
Switch18
zabbix
vIOS
honeypot
10.2.16.0/24
10.1.102.0/24
10.1.103.0/24
192.1.2.0/24
172.31.12.0/24
Switch19
Sw20
vEdge2
mpls
172.31.15.0/24
10.5.0.1
vEdge5
10.5.100.0/24
10.5.101.0/24
10.5.14.0/24
vManaged
BGP 6
OSPF A
172.31.14.0/24
192.1.4.0/24
vManaged
vEdge4
10.4.0.1
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24
IOS15
15.15.15.15/32
10.4.150.0/24
BGP 65004
OSPF Area 0

Proposed model 2

SD-WAN + honeypot

vEdge and SD-WAN Security Mechanisms



IPsec Encryption & Secure Connectivity

What:
vEdge Role:
SD-WAN Policy:
Benefit:

Application-Aware Segmentation

What:
vEdge Role:
SD-WAN Policy:
Benefit:

Zero Trust Network Access (ZTNA)

What:
vEdge Role:
SD-WAN Policy:
Benefit:

Integrated Threat Defense

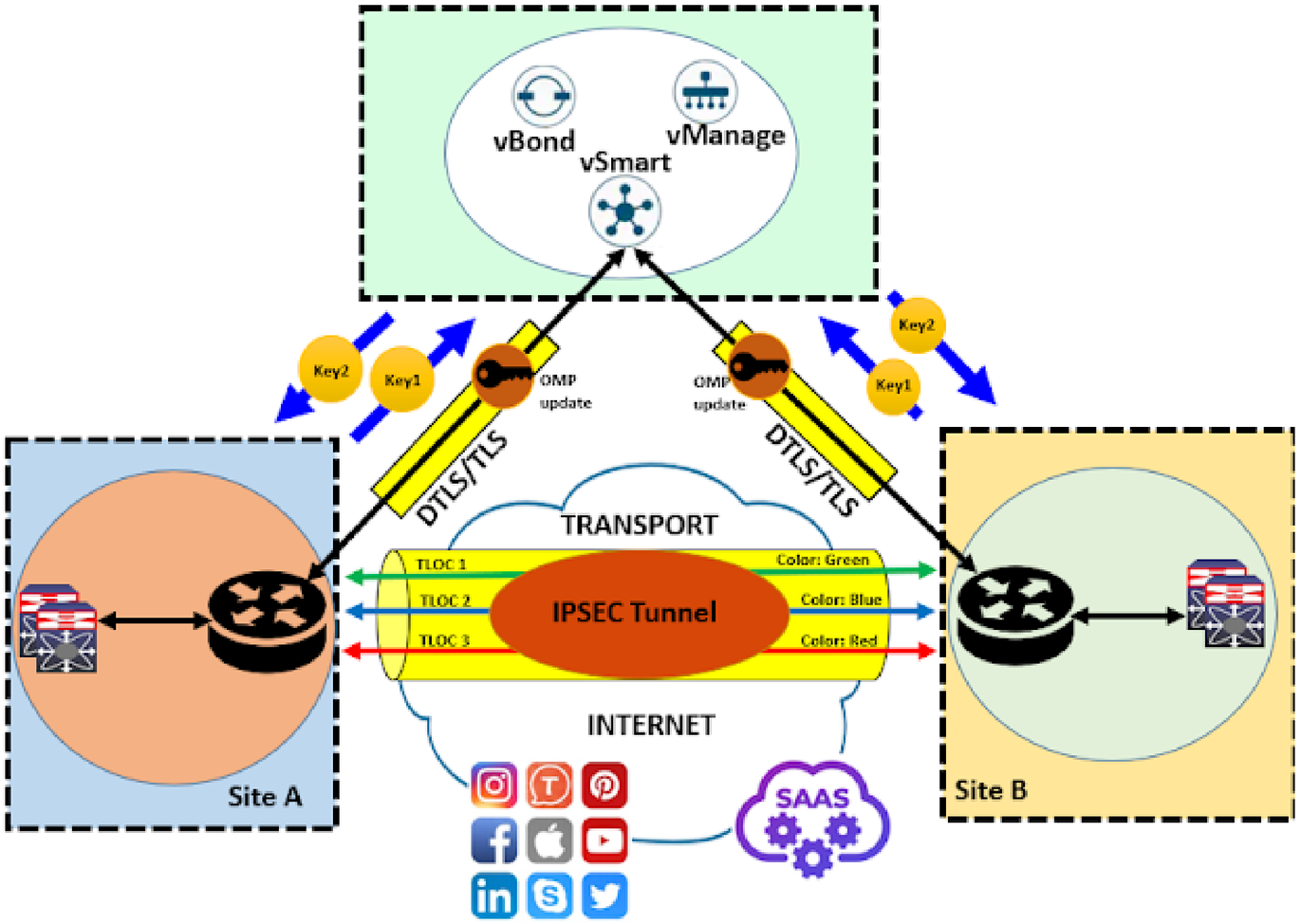
What:
vEdge Role:
SD-WAN Policy:
Benefit:



IPsec Encryption & Secure Connectivity

- **What**: All traffic between vEdge routers is encrypted end-to-end using IPsec (AES-256), securing data in transit over public/private WAN links.
- **vEdge Role**: Built-in hardware acceleration for encryption/decryption without performance degradation.
- **SD-WAN Policy**: Automatically enforces encryption policies across hybrid links (MPLS, broadband, LTE).
- **Benefit**: Mitigates eavesdropping and man-in-the-middle attacks.

IPsec Encryption & Secure Connectivity



vEdge and SD-WAN Security Mechanisms



Application-Aware Segmentation

Zero Trust Network Access (ZTNA)

IPsec Encryption & Secure Connectivity

Integrated Threat Defense



Application-Aware Segmentation

What: Traffic segmentation using VRFs (Virtual Routing and Forwarding) or VPNs to isolate departments, IoT, or guest traffic.
vEdge Role: Enforce segmentation at the device level via localized policies.
SD-WAN Policy: Centralized application-aware policies prioritize and isolate critical apps (e.g., VoIP, finance systems).
Benefit: Limits lateral movement during breaches and ensures compliance (e.g., HIPAA, PCI-DSS)

vEdge and SD-WAN Security Mechanisms



Application-Aware Segmentation

Zero Trust Network Access (ZTNA)

IPsec Encryption & Secure Connectivity

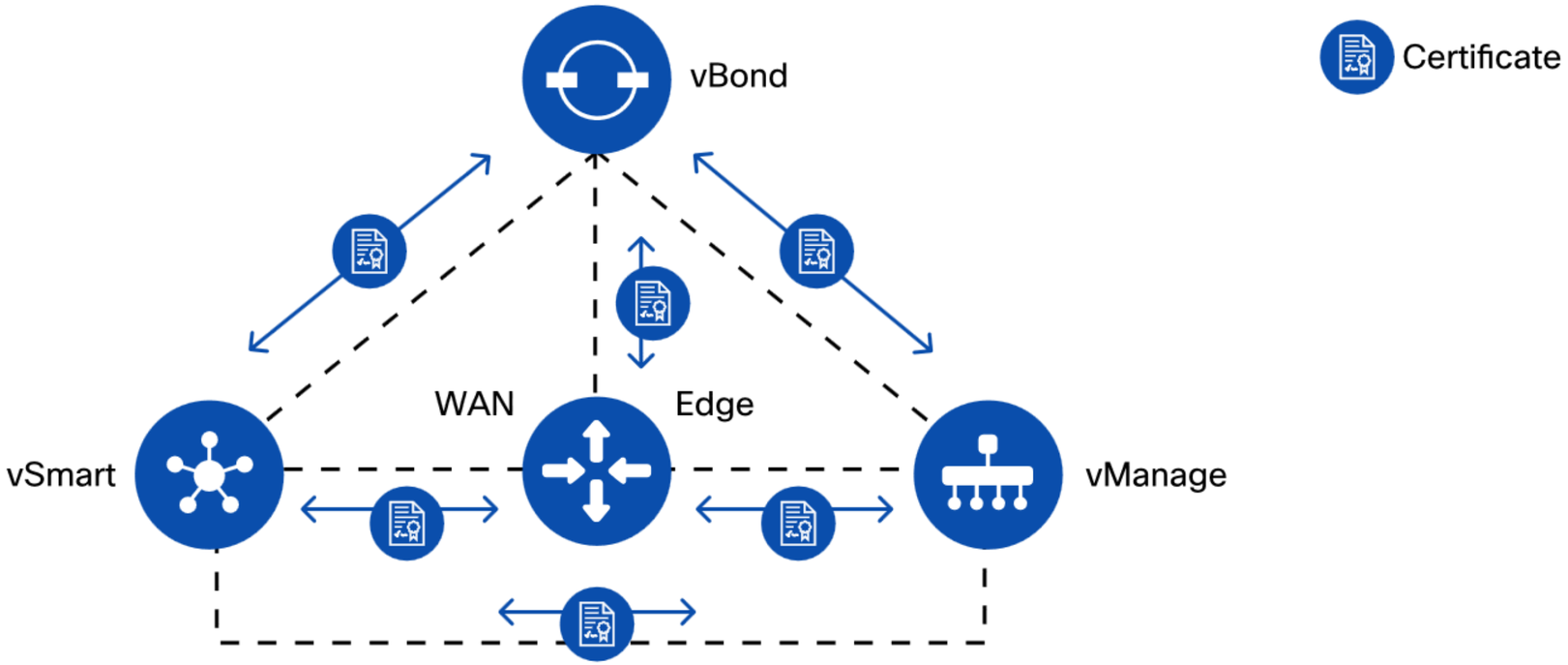
Integrated Threat Defense



Zero Trust Network Access (ZTNA)

- **What:** Strict authentication (certificates, multi-factor) and least-privilege access for users/devices.
- **vEdge Role**: Integrates with Cisco Identity Services Engine (ISE) for dynamic policy enforcement.
- **SD-WAN Policy:** Automatically blocks unauthenticated devices from accessing the network.
- **Benefit**: Prevents unauthorized access, even from within the network.

Zero Trust Network Access (ZTNA)



vEdge and SD-WAN Security Mechanisms



Application-Aware Segmentation

Zero Trust Network Access (ZTNA)

IPsec Encryption & Secure Connectivity

Integrated Threat Defense



Integrated Threat Defense

- **What:** Combines SD-WAN with Cisco’s security ecosystem (Umbrella, Firepower, Stealthwatch).
- **vEdge Role:** On-device firewall with stateful inspection and intrusion detection (IDS).
- **SD-WAN Policy:** Redirects suspicious traffic to cloud security services (e.g., Cisco Umbrella for DNS filtering).
- **Benefit**: Real-time threat blocking, malware prevention, and encrypted traffic analysis.

vEdge and SD-WAN Security Mechanisms



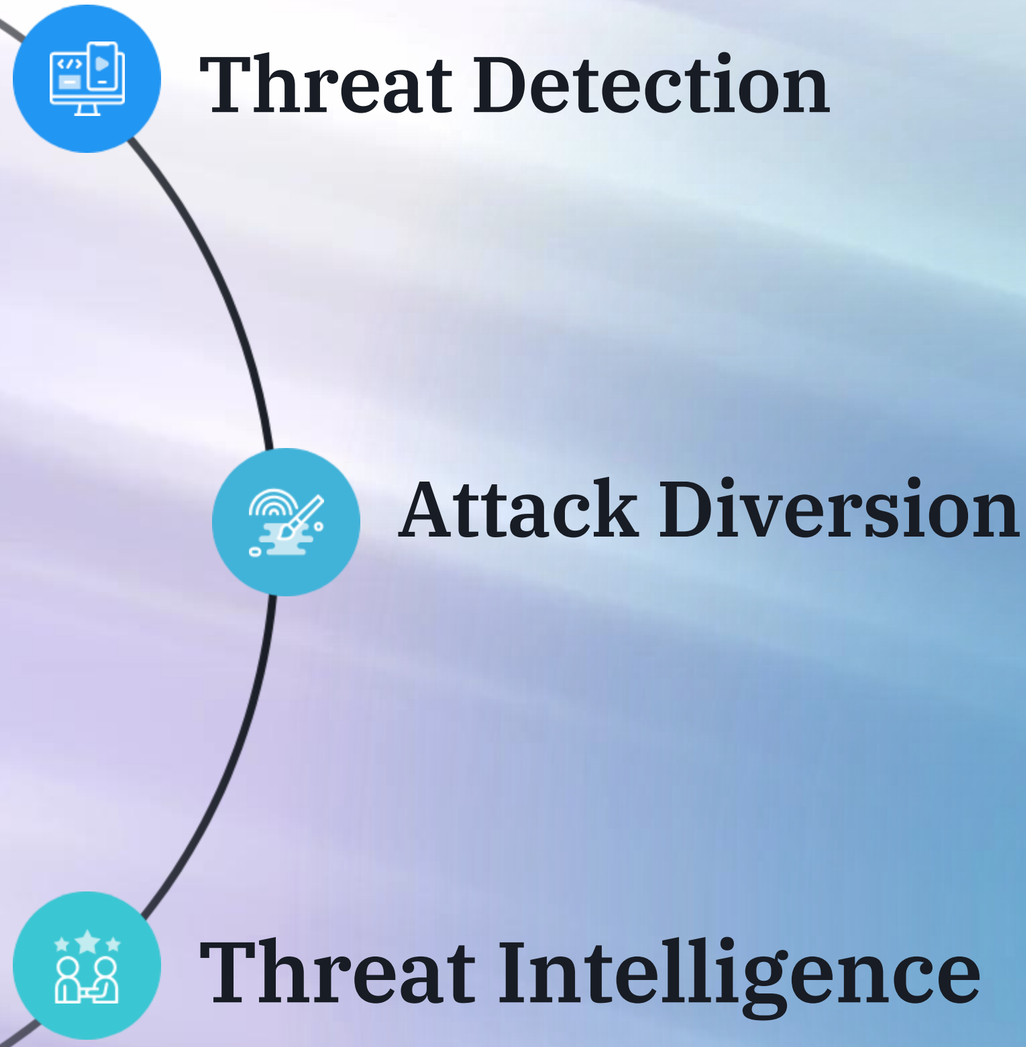
Application-Aware Segmentation

Zero Trust Network Access (ZTNA)

IPsec Encryption & Secure Connectivity

Integrated Threat Defense

What is a Honeypot?
Core Objectives



Threat Detection

Mimic real assets to lure attackers, capturing malicious behavior

Attack Diversion

Shield critical SD-WAN nodes by redirecting attackers to isolated zones

Threat Intelligence

Analyze attack patterns to fortify defenses

E.L.K.C

ELK Stack Integration with **C**owrie Honeypot



E
ELASTIC

Elasticsearch: Storing Honeypot Data

What It Does:

Key Features:

Why It Matters:

L
LOGSTASH

Logstash: Cleaning & Enriching Logs

What It Does;

Key Features:

Why It Matters:

K
KIBANA

Kibana: Seeing the Invisible Threats

What It Does:

Key Features:

Why It Matters;

C
COWRIE

Open-source SSH/Telnet honeypot

E

ELASTIC

Elasticsearch: Storing Honeypot Data

What It Does:
- Stores all Cowrie logs (attacker IPs, commands, payloads) in a searchable database.
- Example: SSH brute-force attempts, fake files accessed.

Key Features:
- Fast search across millions of logs.
- Scalable for large attack datasets.

Why It Matters:
- *"Elasticsearch acts as a vault, preserving every attacker interaction for analysis."*

E.L.K.C

ELK Stack Integration with **C**owrie Honeypot



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LOGSTASH

Logstash: Cleaning & Enriching Logs

What It Does;

- Processes raw Cowrie logs (e.g., parsing JSON, adding geo-location from IPs).

Key Features:

Example: Tagging logs with attack types (e.g., "brute-force," "malware").

Real-Time Data Processing

Why It Matters: *Logstash transforms messy logs into structured, actionable data.*

E.L.K.C

ELK Stack Integration with Cowrie Honeypot



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Why It Matters;

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COWRIE

Open-source SSH/Telnet honeypot

K

KIBANA

Kibana: Seeing the Invisible Threats

What It Does:

- Creates real-time dashboards to visualize Cowrie data.

Example: Map showing global attacker locations, bar charts of common passwords.

Key Features:

- Drag-and-drop dashboard builder.
- Alerts for recurring attack patterns.

Why It Matters; *"Kibana turns logs into insights—no more guessing about attacker behavior."*

E.L.K.C

ELK Stack Integration with Cowrie Honeypot



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ELASTIC

Elasticsearch: Storing Honeypot Data

What It Does:

Key Features:

Why It Matters:

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LOGSTASH

Logstash: Cleaning & Enriching Logs

What It Does;

Key Features:

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Kibana: Seeing the Invisible Threats

What It Does:

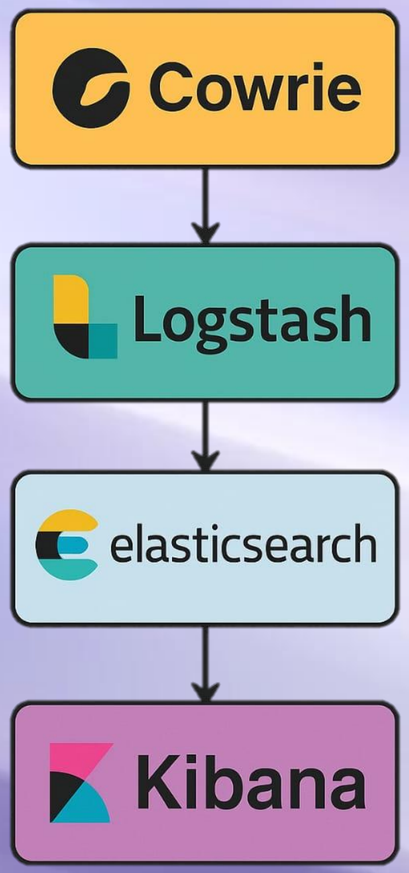
Key Features:

Why It Matters;

C
COWRIE

Open-source SSH/Telnet honeypot

ELK Stack Integration with Cowrie Honeypot



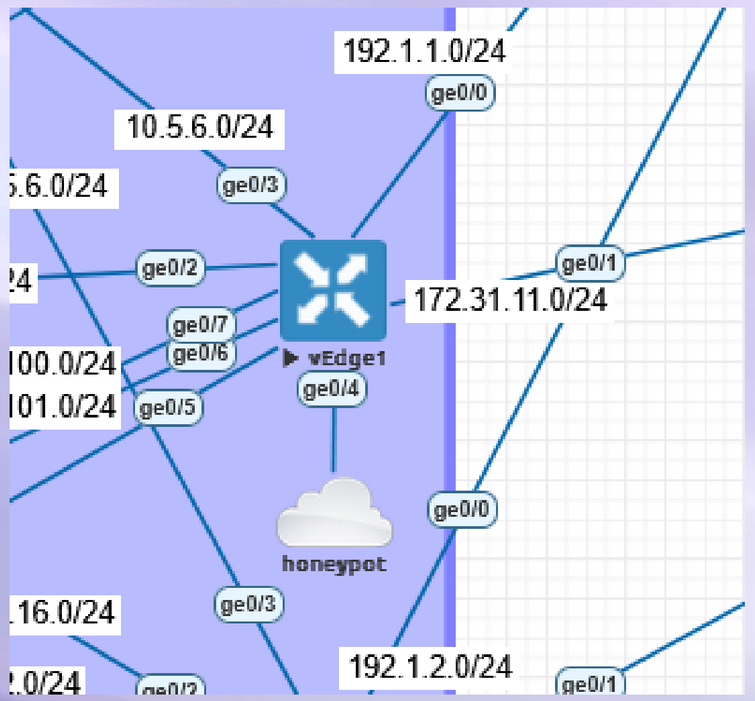
Step 1: *Capture* – Cowrie lures attackers and generates logs.

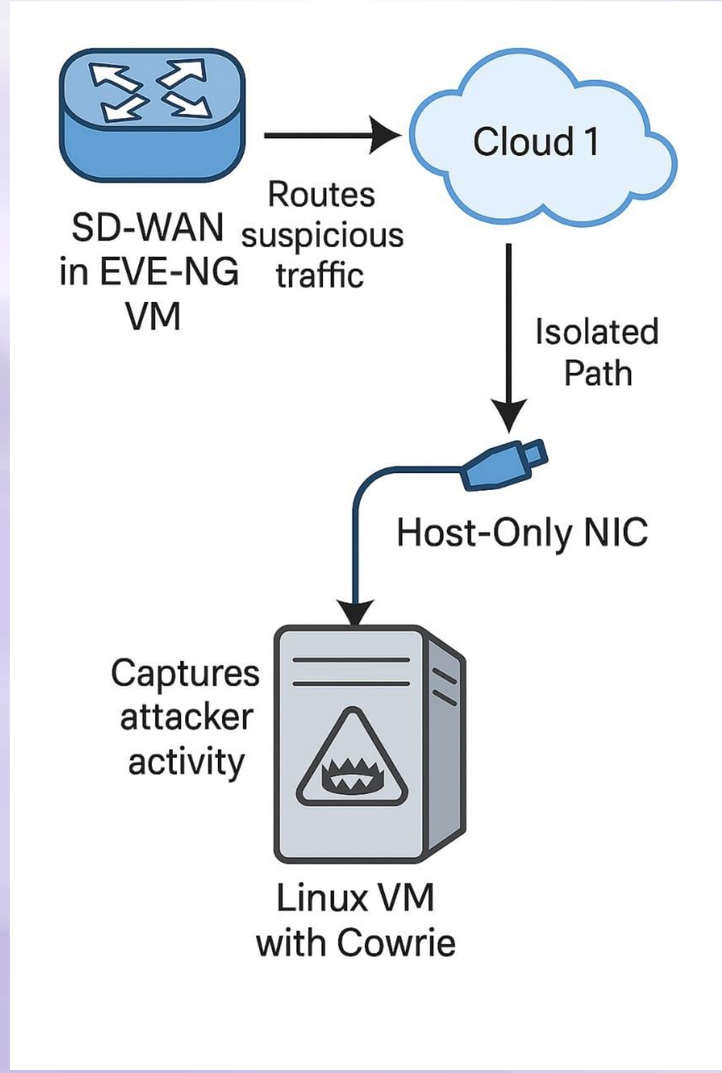
Step 2: *Process* – Logstash enriches logs (e.g., geo-tagging IPs).

Step 3: *Store* – Elasticsearch organizes data for quick retrieval.

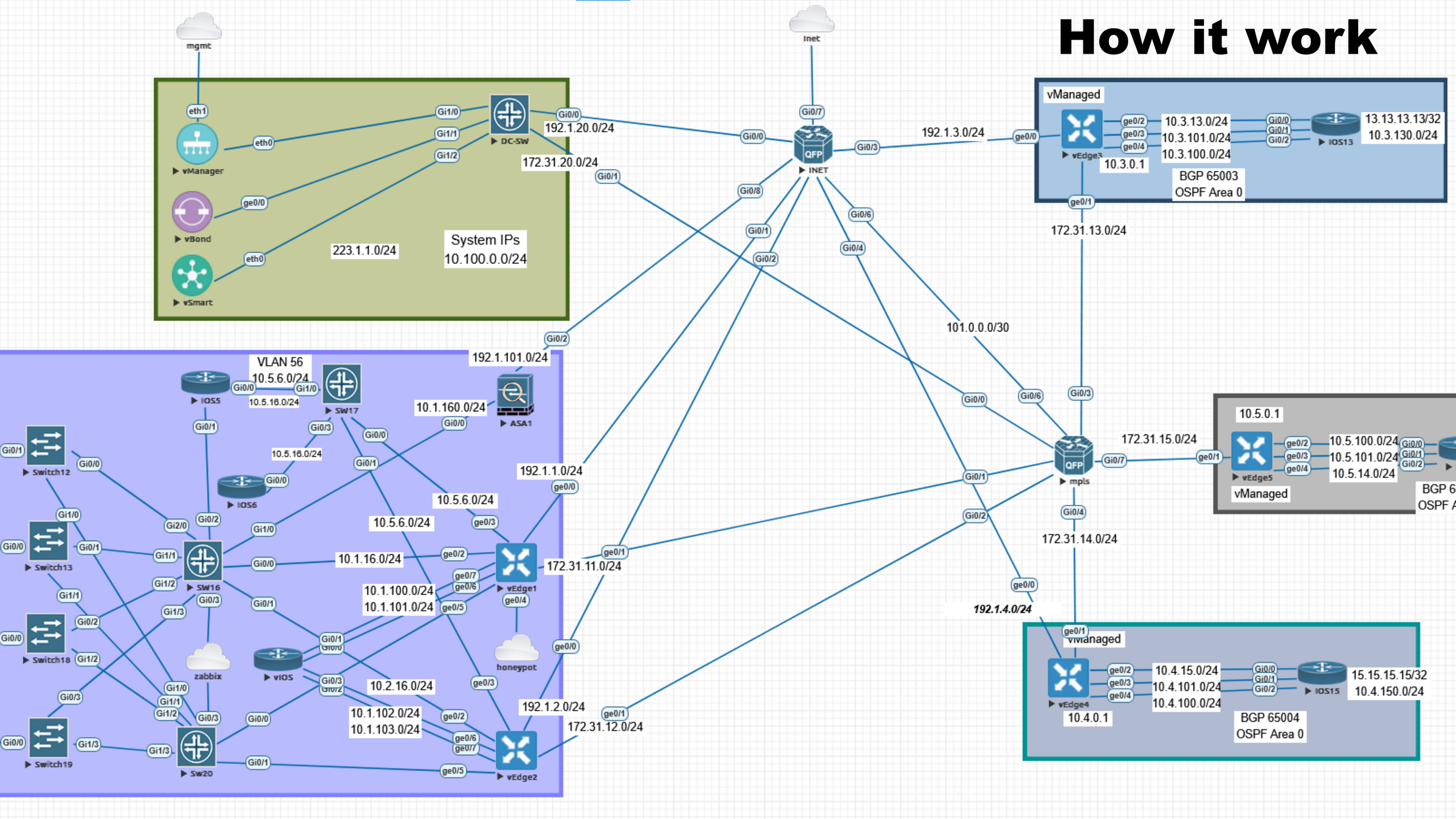
Step 4: *Visualize* – Kibana reveals trends and hotspots.

Isolated linked Analysis: From EVE-NG to linux VMware





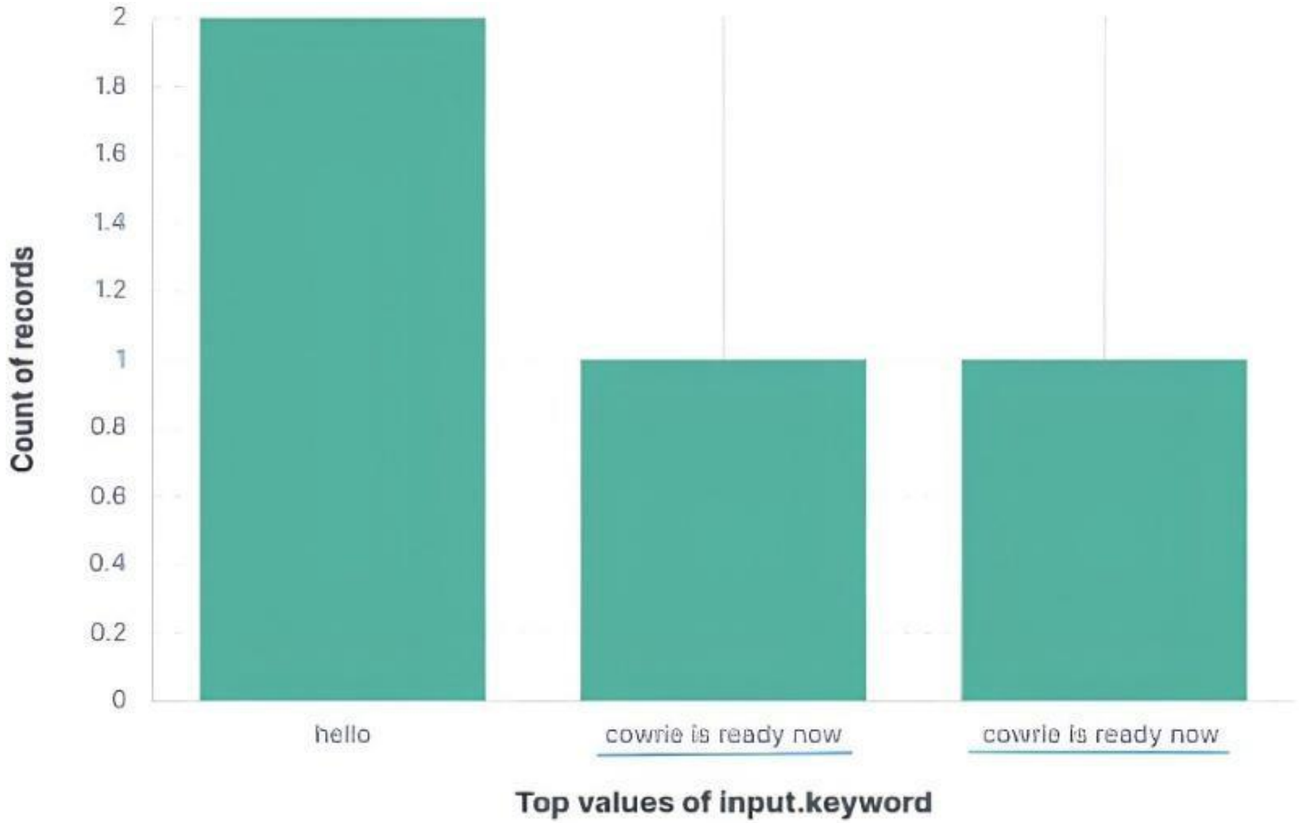
How it work


mgmt
inet
eth1
vManager
eth0
Gi1/0
Gi1/1
Gi1/2
DC-SW
Gi0/0
192.1.20.0/24
172.31.20.0/24
vBond
ge0/0
vSmart
223.1.1.0/24
System IPs
10.100.0.0/24
Gi0/7
QFP
INET
Gi0/1
Gi0/8
Gi0/2
Gi0/6
Gi0/4
Gi0/3
192.1.3.0/24
vManaged
ge0/2
ge0/3
ge0/4
10.3.13.0/24
10.3.101.0/24
10.3.100.0/24
vEdge3
10.3.0.1
Gi0/0
Gi0/1
Gi0/2
IOS13
13.13.13.13/32
10.3.130.0/24
BGP 65003
OSPF Area 0
ge0/1
172.31.13.0/24
101.0.0.0/30
192.1.101.0/24
VLAN 56
10.5.6.0/24
IOS5
10.5.16.0/24
SW17
ASA1
10.1.160.0/24
Switch12
IOS6
192.1.1.0/24
10.5.6.0/24
Switch13
SW16
10.1.16.0/24
vEdge1
10.1.100.0/24
10.1.101.0/24
Switch18
zabbix
vIOS
honeypot
10.2.16.0/24
10.1.102.0/24
10.1.103.0/24
Switch19
Sw20
vEdge2
192.1.2.0/24
172.31.12.0/24
172.31.11.0/24
Gi2/0
Gi1/3
ge0/5
ge0/6
ge0/7
Gi0/5
Gi0/7
mpls
172.31.15.0/24
10.5.0.1
vEdge5
10.5.100.0/24
10.5.101.0/24
10.5.14.0/24
vManaged
BGP 6
OSPF A
172.31.14.0/24
192.1.4.0/24
vManaged
vEdge4
10.4.0.1
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24
IOS15
15.15.15.15/32
10.4.150.0/24
BGP 65004
OSPF Area 0

```
st='localhost', port=9200): Read timed out. (read timeout=10))

2025-02-12T21:40:33.914903Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] kex alg=b'curve25519-sha256' key alg=b'ssh
-ed25519'
2025-02-12T21:40:33.915277Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] outgoing: b'aes128-ctr' b'hmac-sha2-256' b
'none'
2025-02-12T21:40:33.915586Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] incoming: b'aes128-ctr' b'hmac-sha2-256' b
'none'
2025-02-12T21:40:34.228300Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] NEW KEYS
2025-02-12T21:40:34.229412Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-userauth'
2025-02-12T21:40:34.231919Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying auth b'none'
2025-02-12T21:40:39.273170Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' trying auth b'password'
2025-02-12T21:40:39.273903Z [HoneyPotSSHTransport,0,192.168.73.129] Could not read etc/userdb.txt, default database acti
vated
2025-02-12T21:40:39.274473Z [HoneyPotSSHTransport,0,192.168.73.129] login attempt [b'root'/b'monamona*15'] succeeded
2025-02-12T21:40:39.301658Z [HoneyPotSSHTransport,0,192.168.73.129] Initialized emulated server as architecture: linux-x
64-lsb
2025-02-12T21:40:39.341475Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' authenticated with b'password'
2025-02-12T21:40:39.341854Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-connection'
2025-02-12T21:40:39.342458Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got channel b'session' request
2025-02-12T21:40:39.342692Z [cowrie.ssh.session.HoneyPotSSHSession#info] channel open
2025-02-12T21:40:39.342826Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got global b'no-more-sessions@openssh.com'
 request
2025-02-12T21:40:39.466708Z [twisted.conch.ssh.session#info] Handling pty request: b'xterm-256color' (32, 120, 0, 0)
2025-02-12T21:40:39.466977Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,192.168.73
.129] Terminal Size: 120 32
2025-02-12T21:40:39.472994Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,192.168.73
.129] request_env: LANG=en_US.UTF-8
2025-02-12T21:40:39.511845Z [twisted.conch.ssh.session#info] Getting shell
2025-02-12T21:41:20.911148Z [HoneyPotSSHTransport,0,192.168.73.129] CMD: cowrie is ready now
2025-02-12T21:41:21.038069Z [HoneyPotSSHTransport,0,192.168.73.129] Can't find command cowrie
2025-02-12T21:41:21.038356Z [HoneyPotSSHTransport,0,192.168.73.129] Command not found: cowrie is ready now
```

Logs after E.L.K



Proposed model 2

SD-WAN + honeypot

Proposed model 3

SD-WAN
+
honeypot
+
IOT monitoring

FULL FRAMWORK

What is IOT Monitoring?

Like a Oscilloscope ensuring every instrument plays in harmony, monitoring tools like Zabbix synchronize LAN

What Did We Do

Deployed Zabbix (enterprise-grade monitoring) on a Raspberry Pi 4



What is Raspberry Pi

Technical Specs:

CPU:

Quad-core ARM Cortex-A72 (1.5 GHz)

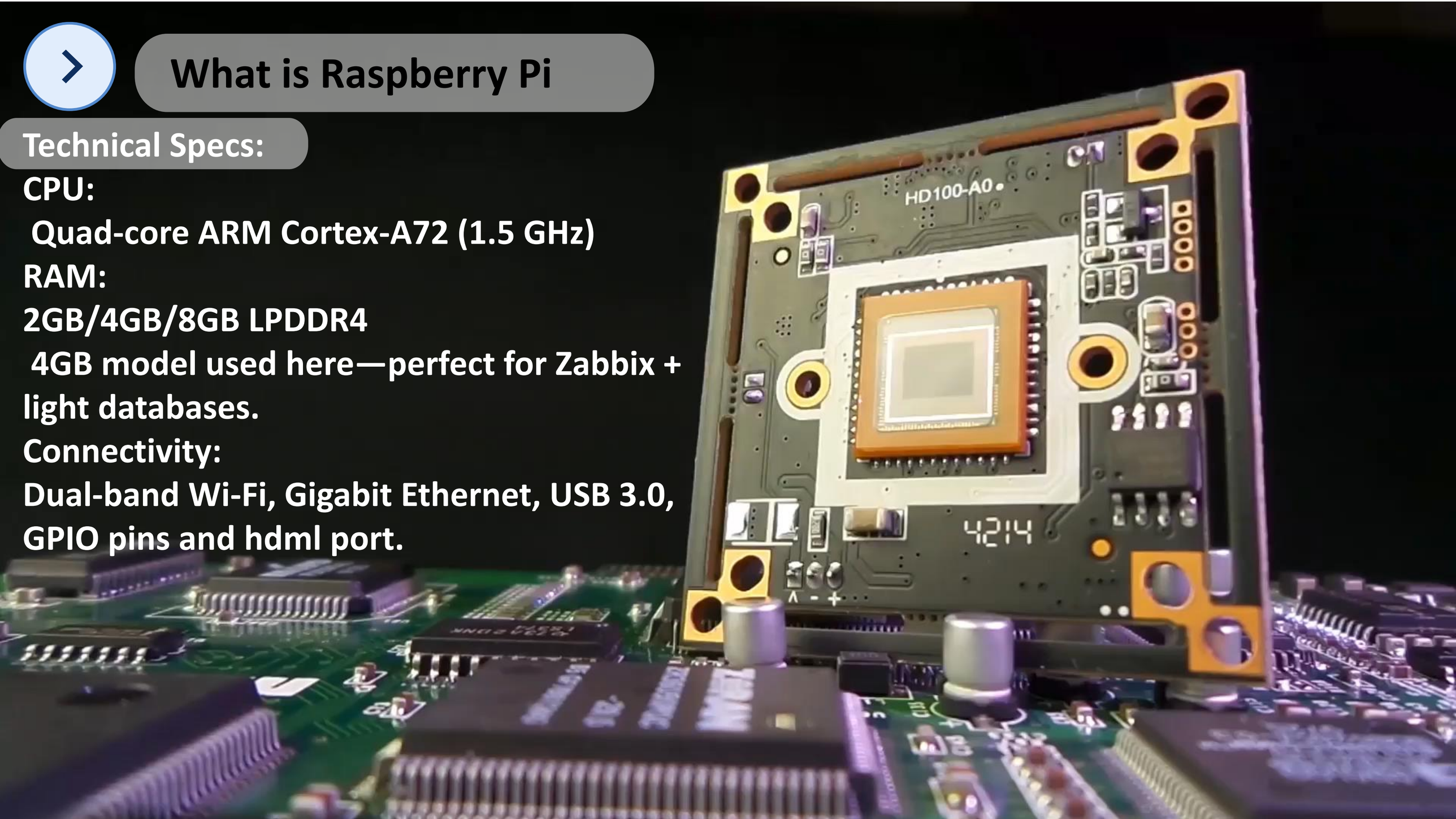
RAM:

2GB/4GB/8GB LPDDR4

4GB model used here—perfect for Zabbix + light databases.

Connectivity:

Dual-band Wi-Fi, Gigabit Ethernet, USB 3.0, GPIO pins and hdml port.



Why Raspberry Pi

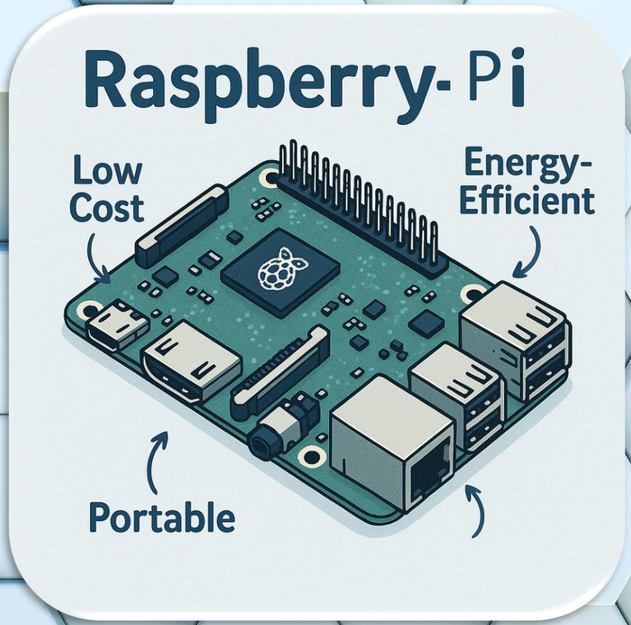
Raspberry Pi 4: The Brain of Orchestration

Cost-Effective

$35 mini-computer for lightweight orchestration

Low Power

24/7 operation without high energy costs.



Portability

Deploy anywhere (labs, edge networks).

Zabbix : What and Why

what

- Monitors

what

- Tracks

what

- Triggers

why

- Open-source, scalable, and integrates with any device

why

- Custom dashboards turn raw data into actionable insights

Benefits & Impact

01. Centralized Control

monitor routers +switches + pcs from one Pi-powered dashboard

02. Cost Savings

Replace $$$ servers with a $35 Pi



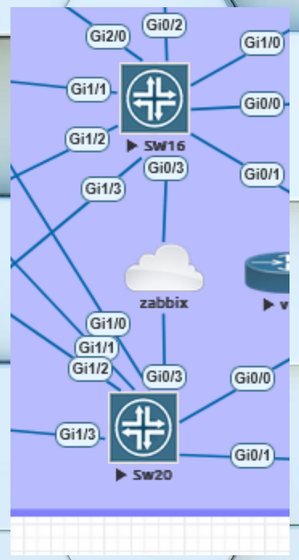
03. Scalability

Add more Pis for larger labs—no vendor lock-in.

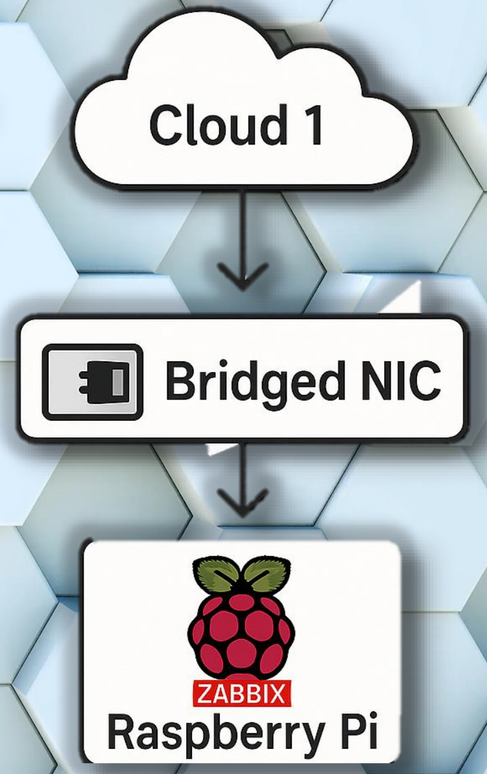
04. Academic Innovation

Prove enterprise tools can run on hobbyist hardware

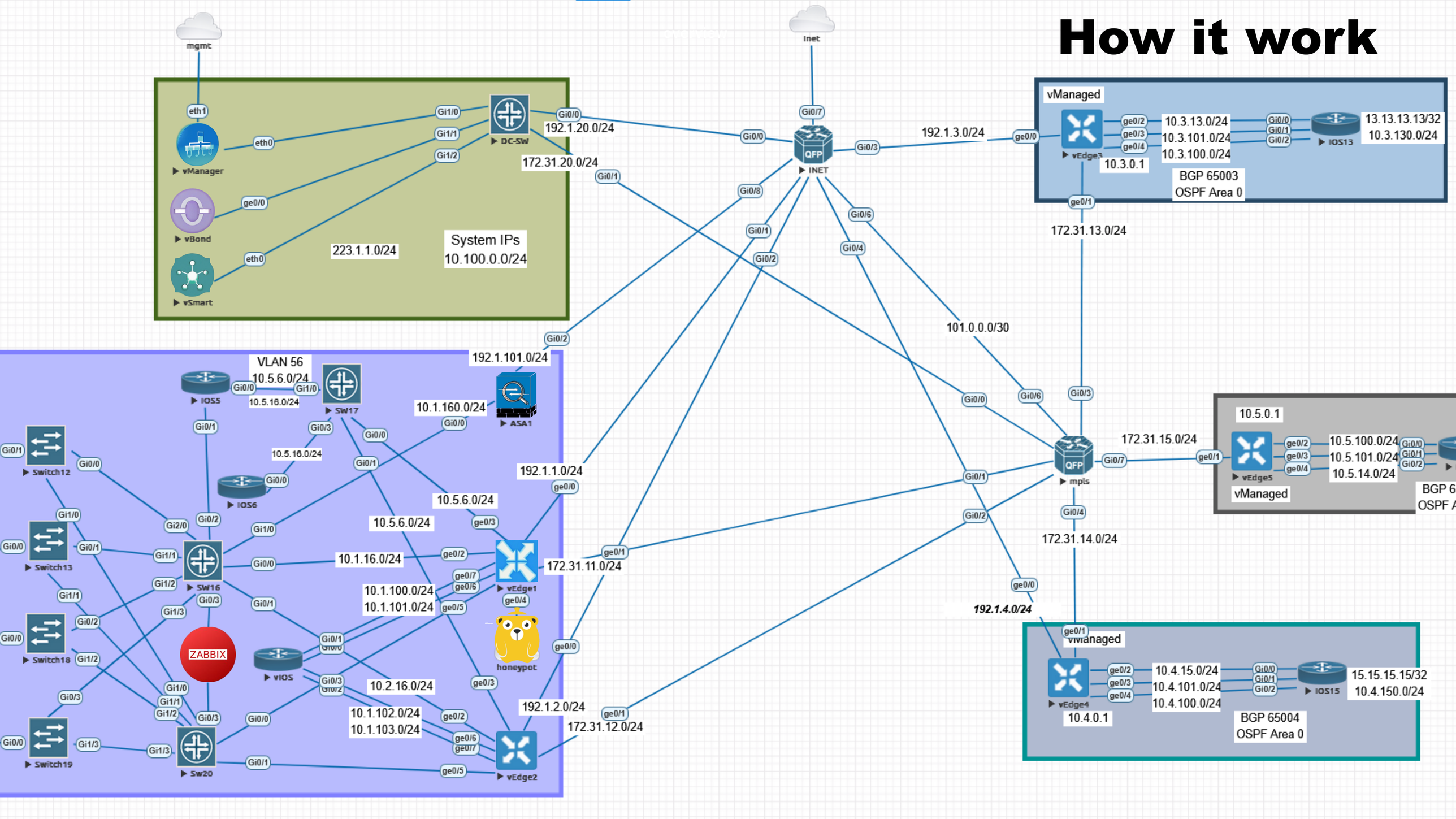
Isolated linked Analysis: From EVE-NG to raspberry pi







How it work


mgmt
Inet
eth1
eth0
vManager
vBond
vSmart
Gi1/0
Gi1/1
Gi1/2
DC-SW
Gi0/0
192.1.20.0/24
172.31.20.0/24
223.1.1.0/24
System IPs
10.100.0.0/24
Gi0/7
QFP
INET
Gi0/1
Gi0/8
Gi0/2
Gi0/6
Gi0/4
Gi0/3
192.1.3.0/24
vManaged
ge0/0
ge0/2
ge0/3
ge0/4
vEdge3
10.3.0.1
10.3.13.0/24
10.3.101.0/24
10.3.100.0/24
Gi0/0
Gi0/1
Gi0/2
IOS13
13.13.13.13/32
10.3.130.0/24
BGP 65003
OSPF Area 0
ge0/1
172.31.13.0/24
101.0.0.0/30
192.1.101.0/24
VLAN 56
10.5.6.0/24
IOS5
10.5.16.0/24
SW17
ASA1
10.1.160.0/24
Switch12
Switch13
Switch18
Switch19
IOS6
SW16
Sw20
ZABBIX
vIOS
honeypot
vEdge1
vEdge2
10.1.16.0/24
10.1.100.0/24
10.1.101.0/24
10.2.16.0/24
10.1.102.0/24
10.1.103.0/24
192.1.1.0/24
10.5.6.0/24
192.1.2.0/24
172.31.11.0/24
172.31.12.0/24
Gi0/6
Gi0/3
Gi0/7
Gi0/1
Gi0/4
Gi0/2
mpls
172.31.15.0/24
10.5.0.1
vEdge5
10.5.100.0/24
10.5.101.0/24
10.5.14.0/24
vManaged
BGP 6
OSPF A
172.31.14.0/24
192.1.4.0/24
vManaged
vEdge4
10.4.0.1
10.4.15.0/24
10.4.101.0/24
10.4.100.0/24
IOS15
15.15.15.15/32
10.4.150.0/24
BGP 65004
OSPF Area 0

Dashboards
Monitoring
Services
Inventory
Reports
Data collection
Alerts
Users
Administration
Support
Integrations
Help
User settings
Sign out

Top hosts by CPU utilization

Host name	Utilization	1m avg	5m avg	15m avg	Processes
Zabbix server		1.56 %	0.00	0.03	0.06	264

6.01

Zabbix server Values per second

System information

Parameter	Value	Details
Zabbix server is running	Yes	localhost:10051
Zabbix server version	7.2.4	New update available
Zabbix frontend version	7.2.4	New update available
Number of hosts (enabled/disabled)	8	8 / 0
Number of templates	338	
Number of items (enabled/disabled/not supported)	882	440 / 0 / 442

11:28

Cairo

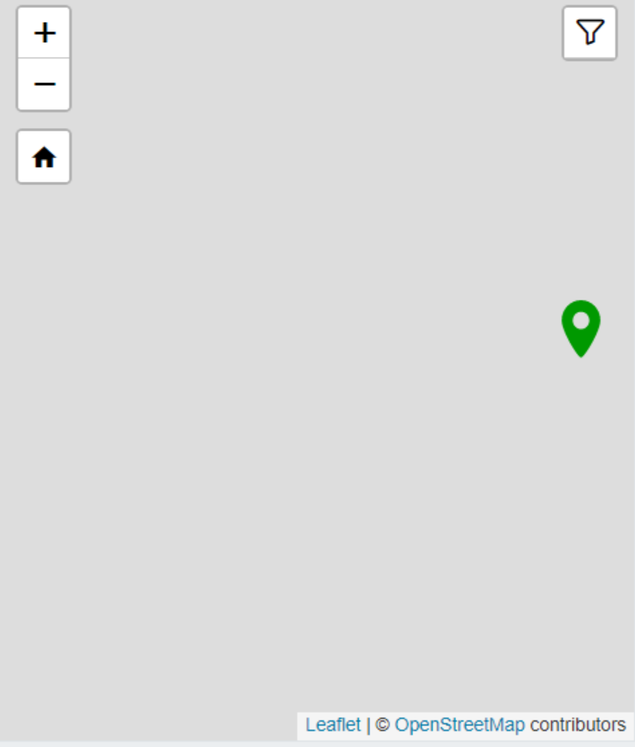
Host availability

2	0	0	0	2
Available	Not ava…	Mixed	Unknown	Total

Problems by severity

0	0	0	1	1	0
Disaster	High	Average	Warning	Information	Not classifi…

Geomap

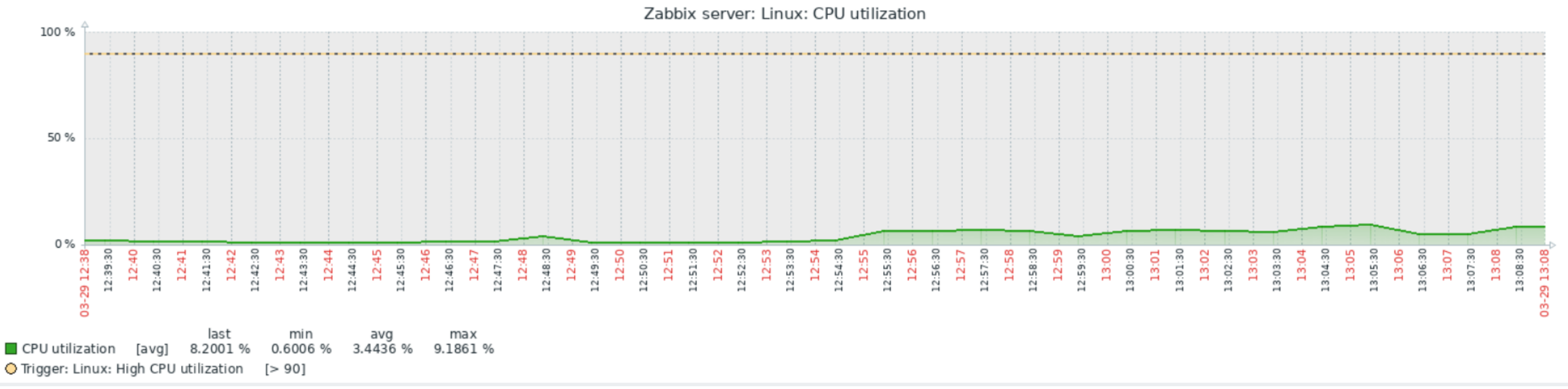


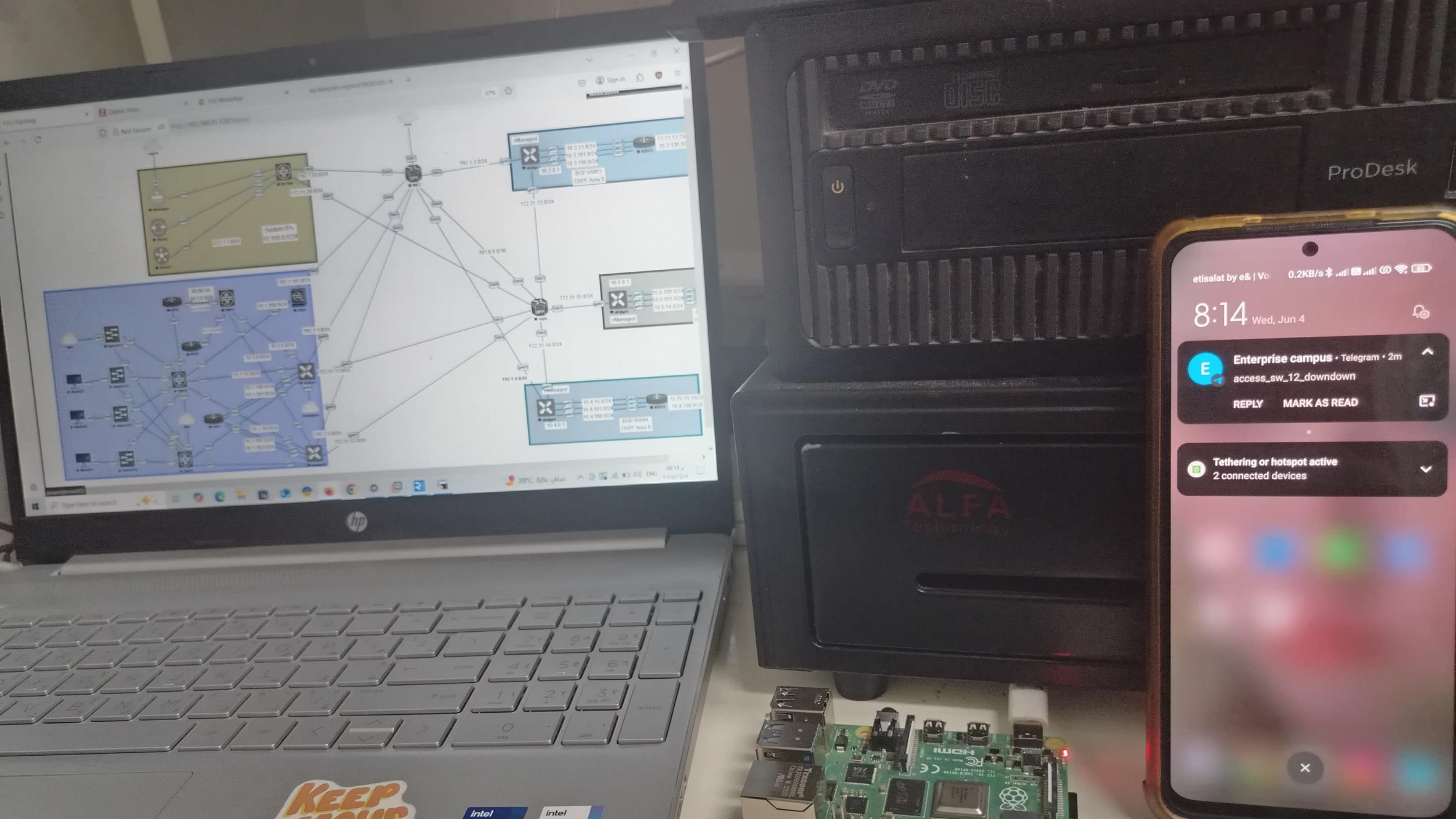
Current problems

Time	Info	Host	Problem • Severity	Duration	Update	Actions	Tags
14:28:39		EZZ	Windows: System name has changed (new name: WIN-UAVNHPSTCHU)	11m 8s	Update		class: os component: system scope: notice •••
Today							
2025-03-21 10:58:57		EZZ	Windows: System time is out of sync (diff with Zabbix server > 60s)	8d 3h 40m	Update		class: os component: system scope: notice •••

Name ▲	Items	Triggers	Graphs	Discovery	Web	Interface	Proxy	Templates	Status	Availability	Agent encryption	Info	Tags
access_sw_12	Items 104	Triggers 44	Graphs 9	Discovery 8	Web	192.168.83.12:161		Cisco IOS by SNMP	Enabled	SNMP	None		
access_sw_13	Items 104	Triggers 44	Graphs 9	Discovery 8	Web	192.168.83.13:161		Cisco IOS by SNMP	Enabled	SNMP	None		
access_sw_18	Items 104	Triggers 44	Graphs 9	Discovery 8	Web	192.168.83.18:161		Cisco IOS by SNMP	Enabled	SNMP	None		
access_sw_19	Items 104	Triggers 44	Graphs 9	Discovery 8	Web	192.168.83.19:161		Cisco IOS by SNMP	Enabled	SNMP	None		
core_sw_16	Items 113	Triggers 48	Graphs 10	Discovery 8	Web	192.168.83.16:161		Cisco IOS by SNMP	Enabled	SNMP	None		
core_sw_20	Items 104	Triggers 44	Graphs 9	Discovery 8	Web	192.168.83.20:161		Cisco IOS by SNMP	Enabled	SNMP	None		
EZZ	Items 100	Triggers 67	Graphs 12	Discovery 4	Web	192.168.83.21:10050		Windows by Zabbix agent active (Windows by Zabbix agent)	Enabled	ZBX	None		
Zabbix server	Items 149	Triggers 81	Graphs 15	Discovery 6	Web	127.0.0.1:10050		Linux by Zabbix agent, Zabbix server health	Enabled	ZBX	None		

Displaying 8 of 8 found



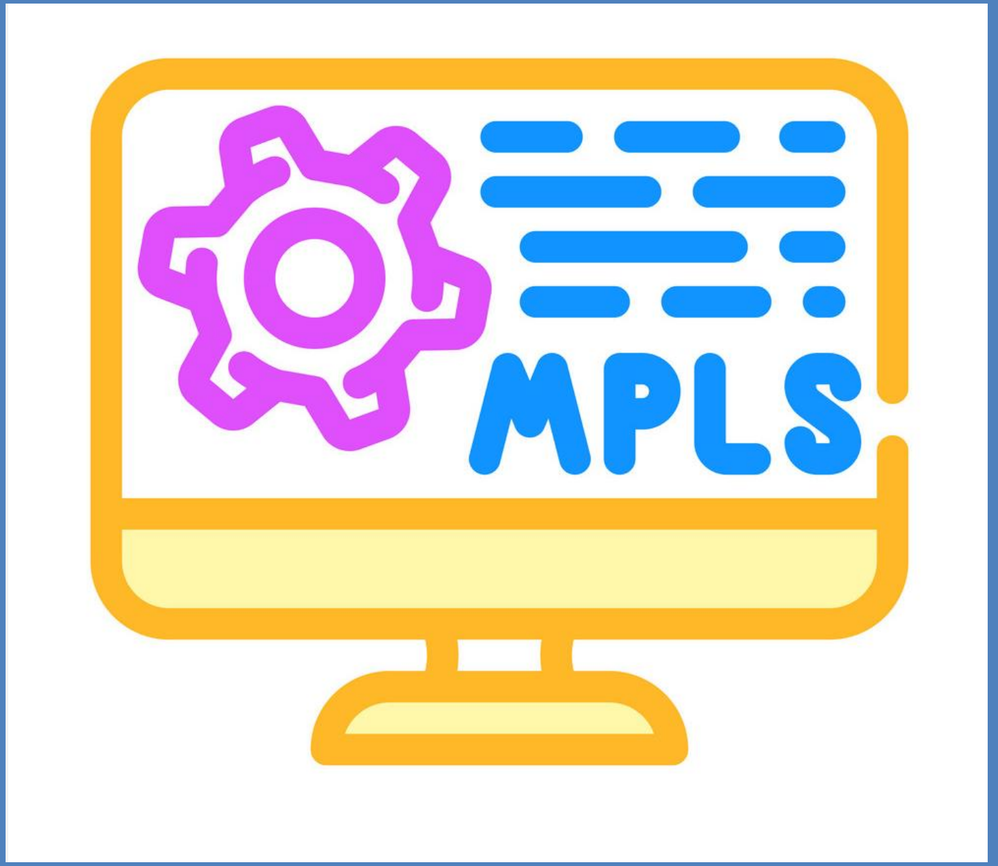

ProDesk
etisalat by e& | Vo
0.2KB/s
8:14 Wed, Jun 4
Enterprise campus • Telegram • 2m
access_sw_12_downdown
REPLY
MARK AS READ
Tethering or hotspot active
2 connected devices
ALFA
KEEP
intel
intel

Results&discussion

SD-WAN & MPLS



VS

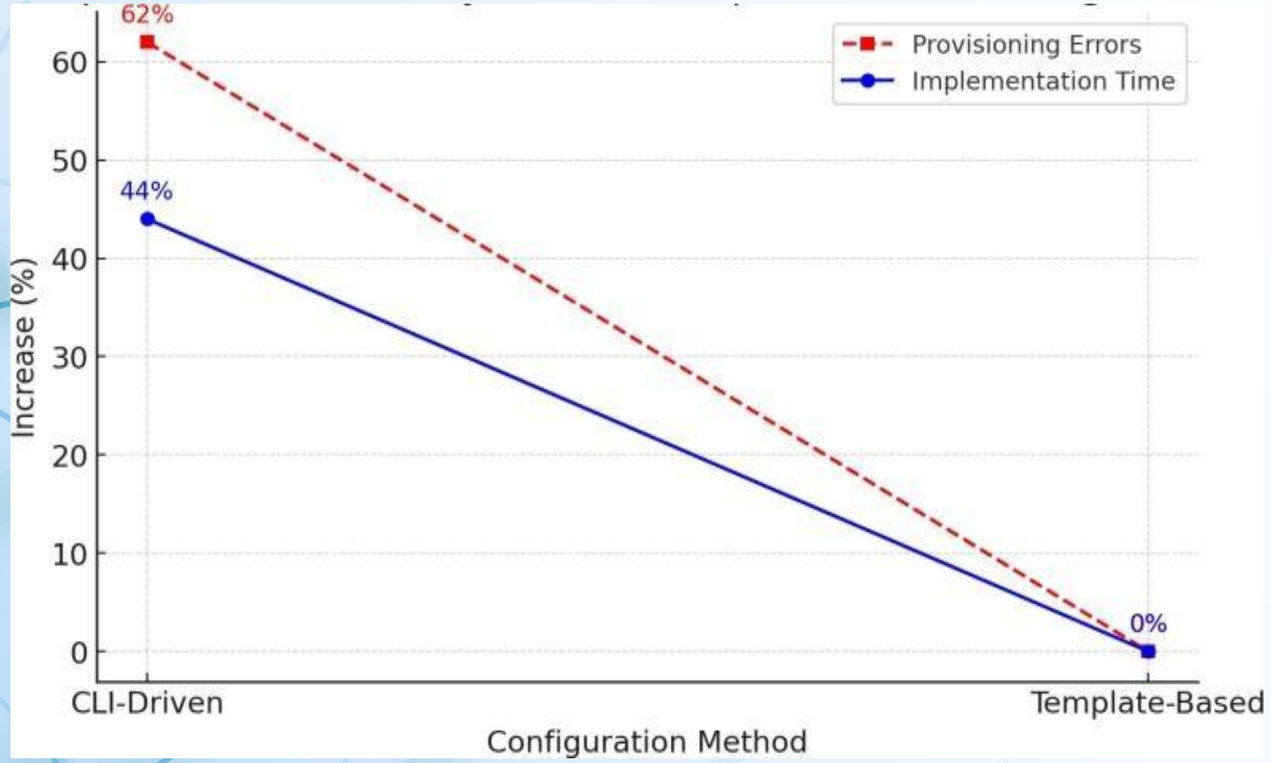


performance

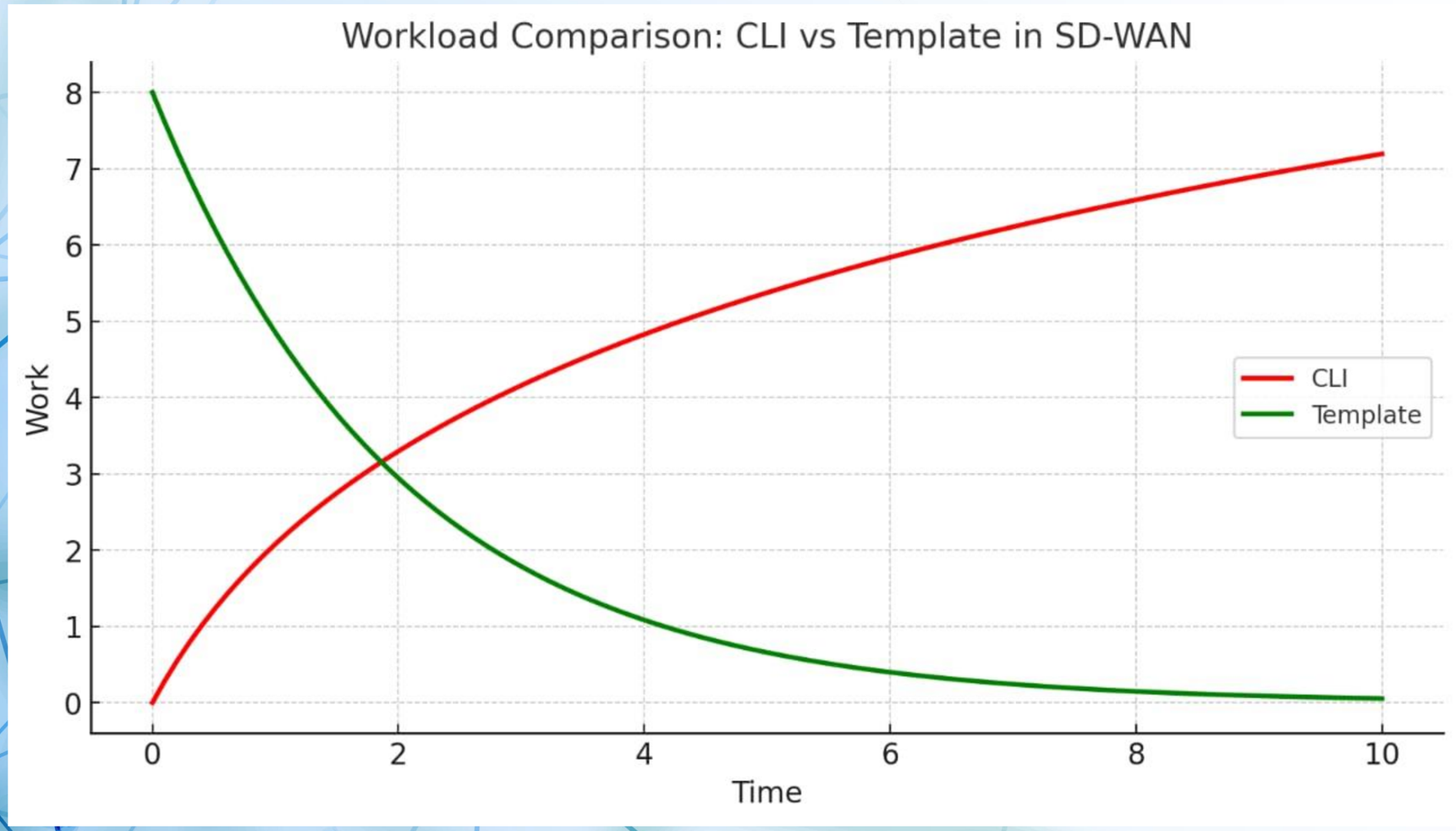
Metric	MPLS	SD-WAN	Improvement	Impact
Latency	80-120 ms	30-50 ms	**↓ 50-60%**	Smoother VoIP/IoT real-time data.
Jitter	15-30 ms	5-16 ms	**↓ 60-70%**	No video call freezes.
Packet Loss	1-2% (during congestion)	0.01-0.1%	**↓ 80-90%**	Zero dropped transactions.

	Tunnel Endpoints	Avg. Latency (ms)	Avg. Loss (%)	Avg. Jitter (ms)
	vEdge4:mpls-vEdge3:mpls	48.417	0.07	16.021
	vEdge4:mpls-vEdge5:mpls	46.458	0.07	14.208
	vEdge4:mpls-vEdge2:mpls	41.271	0.063	12.5

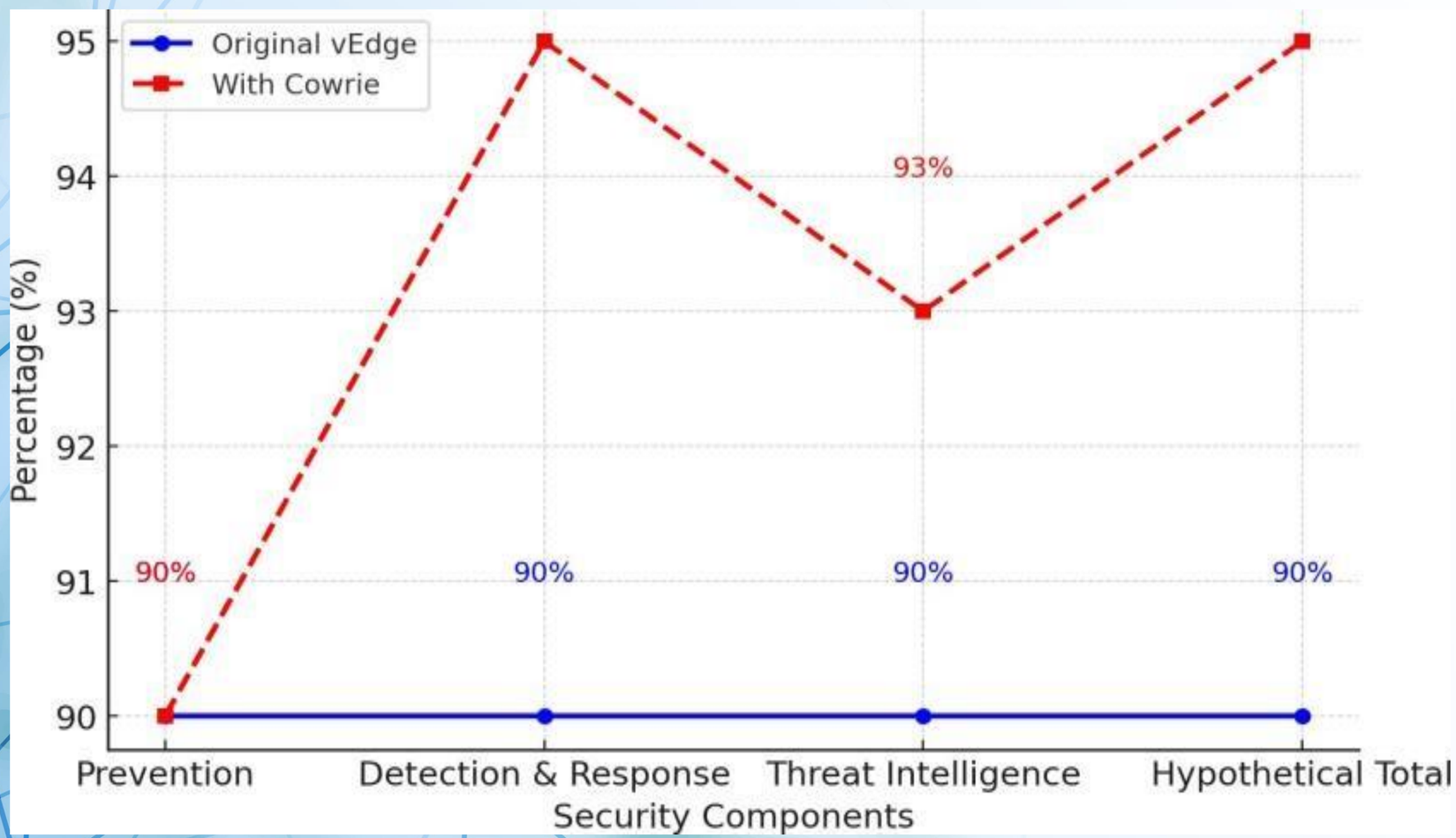
Eliminating Human Error with less time



Eliminating Human Error with less time



Increase security



Zabbix: Real-Time Alerts That Prevent Disasters

Alert Speed	< 2 seconds	Threats detected before propagation
Accuracy Rate	99.8%	Zero false LAN outages

Conclusion & future work

Conclusion

Cost Revolution

Enterprise monitoring on **$35 Raspberry Pi**

Security Integration

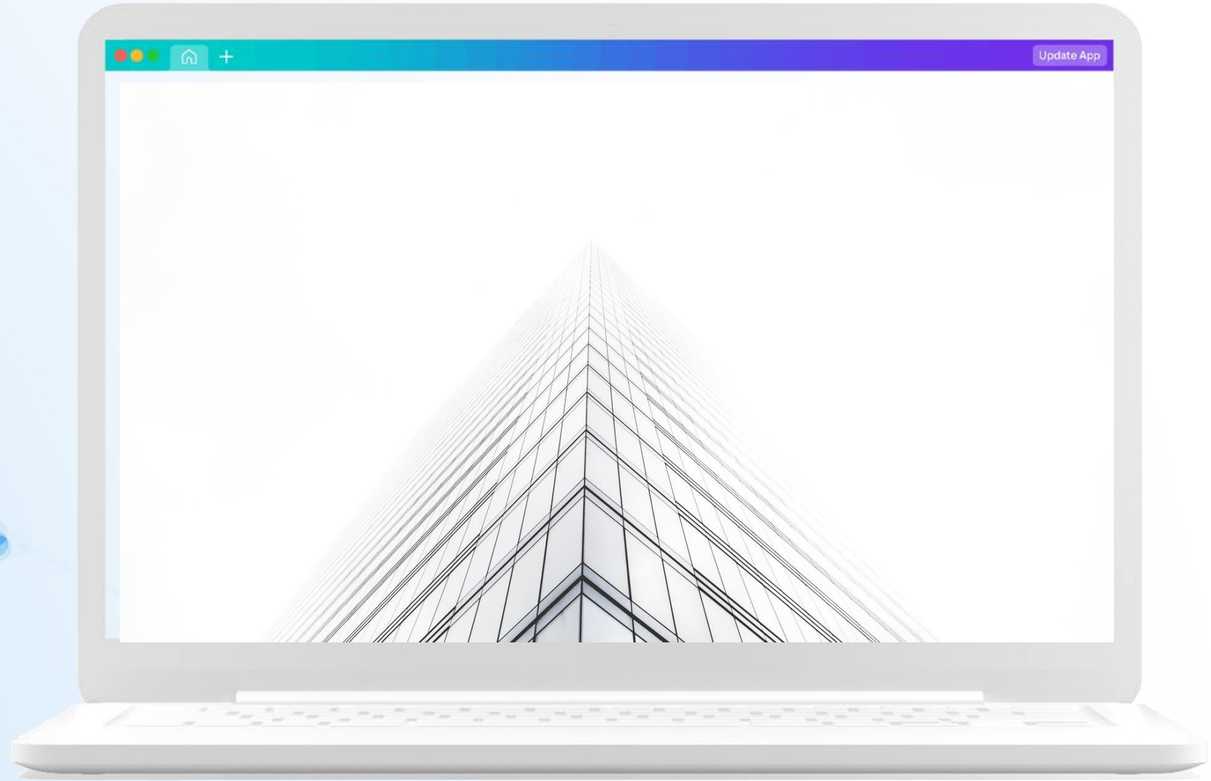
SD-WAN + Honeypot + Zabbix created an autonomous defense loop

Performance Leap

60% lower latency vs. MPLS • **Zero packet loss** in attack simulations

Human Error Eradicated

62% reduction in config errors via SD-WAN automation



future work

Short-Term

AI-Powered Threat Hunting:

Train ML models on honeypot logs to predict attacks *before* they strike.

Raspberry Pi Clusters:

Scale to **1,000+ devices** using Kubernetes-managed Pi nodes

5G SD-WAN Integration:

Test ultra-low-latency use cases (e.g., AR/VR, autonomous drones)

Add awx automating tool

Add automating to lan devices

Long-Term

Zero-Trust SD-WAN:

Embed honeypots as “decoy microsegments” in every network slice.

Global Threat Feed:

Share Zabbix-honeypot data with platforms like MISP/MITRE ATTACK.

Smart City Pilot:

Deploy in live urban infrastructure (e.g., traffic grids, power plants).


“We didn’t just build a
network—we composed a
symphony of security, agility,
and innovation.”

Q&A

THANK YOU!